

Direction-Dependent Band Gaps and Wave Steering in Two-Dimensional Strain-Gradient Elastic Media with Anisotropic Ellipsoidal Microstructure: A Bloch–Floquet C^1 Finite Element Analysis

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Haryana State Research Fund Project No. 14/66-2025 SPD/HSHEC/HSRF, Session 2025–2026

September 2026

Abstract

Microstructured solids exhibit rich dispersive phenomena when acoustic wavelengths approach characteristic internal length scales, where classical local elasticity breaks down. While prior investigations of strain-gradient metamaterials have largely been restricted to one-dimensional chains or two-dimensional media with isotropic length scales, the role of microstructural orientation as an independent design knob remains unaddressed. In this work, we present a two-dimensional Bloch–Floquet finite element framework for Mindlin Form-II strain-gradient elasticity incorporating an anisotropic characteristic-length tensor derived from the second moments of a rotated ellipsoidal averaging domain, alongside an independent micro-inertia parameter ℓ_i . A C^1 -conforming Bogner–Fox–Schmit (BFS) 32-degree-of-freedom rectangular element is formulated, wherein complex Bloch phase transformations are consistently applied across nodal displacements as well as first-order and mixed-second-order derivative degrees of freedom. The numerical implementation is verified against an eight-test internal consistency suite and mesh convergence across 4^2 to 32^2 elements, yielding an observed empirical least-squares convergence rate of $p = 4.17$ (fitted over the refinement levels admissible under a pre-declared measurement-resolution rule; 95% CI: [3.15, 5.20]; no theoretical order claimed) and a locked operational numerical resolution floor of $\varepsilon_\Delta = 4.63 \times 10^{-11}$. In a homogeneous microstructured medium (Case H), we demonstrate that microstructural anisotropy induces strong directional frequency migration along Γ – X while strictly preserving the absence of complete omnidirectional band gaps ($\Delta_{\text{complete}} \leq -0.3235$). The resulting (θ, AR) design map reveals an orientation sensitivity metric S_θ that is largest at the most elongated configuration (2.34 rad^{-1} at $\text{AR} = 10$), accompanied by an acoustic wave-vector steering deviation of up to $\delta_{\text{max}} = 2.78^\circ$ at $\bar{k} = 0.5$ (invariant to within 10^{-5} deg over the full orientation grid $\theta \in [0^\circ, 90^\circ]$ at $\text{AR} = 10$, with the steering figure of merit $M_s = \phi^*(90^\circ) - \phi^*(0^\circ)$ equal to 0° at $\text{AR} = 5$ and 10° at $\text{AR} = 10$ on the locked 5° sampling grid). Finally, asymptotic analysis and numerical evidence demonstrate that micro-inertia is constitutively necessary to bound high-wavenumber phase velocities to a finite horizon $v_{T,\infty} = 0.3162$, ensuring physical admissibility.

Highlights:

- Bloch–Floquet C^1 FE framework for 2D anisotropic strain-gradient elasticity.
- Rotated ellipsoidal neighborhood yields independent orientation θ and aspect ratio AR.
- Consistent Bloch phase transformations applied to value and derivative DOFs.
- Micro-inertia is proved constitutively necessary to bound short-wave speeds.
- Directional stop bands and wave steering mapped over (θ, AR) design space.

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Keywords: Strain-gradient elasticity; Mindlin Form-II; Micro-inertia; Bloch–Floquet analysis; C^1 Bogner–Fox–Schmit element; Anisotropic length tensor; Directional band gaps; Wave steering.

1 Introduction

1.1 Microstructured Solids and Length-Scale Effects

The propagation of elastic waves through structured and architected solids has emerged as a cornerstone of modern mechanics, acoustic metamaterial design, and vibration isolation [1]. In classical Cauchy elasticity, stress is assumed to depend strictly on the local strain tensor, implying an inherently scale-free continuum without an internal characteristic length. However, when the acoustic or elastic wavelength approaches the characteristic dimensions of the underlying microstructural architecture—such as the grain size in textured polycrystals, the fiber spacing in advanced composites, or the cell pitch in additively manufactured lattices—classical continuum mechanics ceases to describe observed dispersive and size-dependent behaviors [2, 3]. Under such short-wave conditions, non-local strain interactions and higher-order deformation gradients become dominant, necessitating generalized continuum theories.

1.2 Gradient Elasticity and the Necessity of Micro-Inertia

Among higher-order continuum representations, Mindlin’s strain-gradient theory [2, 4] provides a rigorous variational framework incorporating displacement gradients into the strain energy density. In Mindlin’s Form-II formulation, the strain energy is parameterized by classical Lamé constants alongside higher-order characteristic length scales that penalize strain gradients [5, 6]. While static strain-gradient elasticity successfully captures boundary-layer stiffening, fracture regularisation, and micro-scale size effects, dynamic formulations encounter fundamental physical requirements. Specifically, if higher-order gradient elasticity is formulated without higher-order kinetic energy (i.e., with classical kinetic energy alone), the resulting dispersion relation yields phase and group velocities that grow unbounded as wavenumber $k \rightarrow \infty$ ($v_p \propto k$). As proven asymptotically in Appendix A, such unbounded high-wavenumber phase velocities violate causality and lead to physically inadmissible short-wave responses.

To overcome this pathology, Mindlin introduced the concept of micro-inertia, which supplements the classical kinetic energy density with terms proportional to the velocity gradients [2]. The micro-inertia parameter, denoted here by ℓ_i , acts as a dynamic length scale that balances the higher-order spatial stiffness operators. As established in 1D analyses by Papargyri-Beskou & Beskos [7] and Li et al. [8], non-zero micro-inertia bounds the high-wavenumber phase velocity to a finite acoustic limit, ensuring a physically admissible continuum representation across the entire spectrum.

1.3 Phononic Crystals and Direction-Dependent Band Gaps

In periodic elastodynamic media, spatial modulation of constitutive properties leads to band structures characterized by passing pass bands and destructive interference stop bands (band gaps) governed by the Floquet–Bloch theorem [1]. In classical composites, band gaps arise predominantly through periodic impedance contrast (Bragg scattering) or localized resonances. In contrast, higher-order gradient media exhibit intrinsic microstructural dispersion even in the absence of material heterogeneity. Furthermore, when microstructural length scales exhibit directional preference—such as elongated grains from rolling or oriented micro-fibers—the effective non-local interactions become anisotropic [9].

1.4 Literature Positioning and State of the Art

A comprehensive survey of published wave-dispersion and band-gap investigations based on gradient elasticity reveals that the literature has been overwhelmingly confined to one-dimensional systems or two-dimensional media with isotropic characteristic lengths. Table 1 provides a systematic positioning of the present study relative to foundational and recent contributions.

As shown in Table 1, early 1D analytical and transfer-matrix studies by Zheng & Wei [10], Papargyri-Beskou & Beskos [7], and Li, Wei & Zhou [8] explored basic microstructural dispersion and partial gaps in layered media. In 2D media, Zhan & Wei [9] investigated anisotropic band gaps

Table 1: Positioning of the present study in the context of published gradient-elastic elastodynamic investigations.

Reference	Year	Dim.	Continuum theory	Micro-inertia	Anisotropic l	Orientation θ	Numerical method	Reported wave quantities
Zheng & Wei (2009)	2009	1D	Grad. elast.	No	No	No	Transfer matrix	Dispersion relation
Papargyri-Beskou & Beskos (2009)	2009	1D	Grad. elast.	Yes	No	No	Analytic	Dispersion relation
Zhan & Wei (2010)	2010	2D	Classical	No	Yes (aniso)	Yes	Plane-wave exp.	Band gaps
Li, Wei & Zhou (2016)	2016	1D	Grad. elast.	Yes	No	No	Transfer matrix	Dispersion, partial gaps
Hosseini & Zhang (2021)	2021	2D	Grad. elast.	No	No	No	Meshless radial PIM	Band gaps (isotropic)
Li et al. (2023)	2023	1D	Grad. thermoelast.	Yes	No	No	Transfer matrix	Dispersion, partial gaps
[Anchor A] Li et al. (2024)	2024	1D	Flexo. + grad.	Yes	No	No	Transfer matrix	Dispersion, partial gaps
[Anchor B] Mishra et al. (2026)	2026	2D	Grad. lattice	No	No	No	Dyn. stiffness / FB	Dispersion relation
[Anchor C] Present study	2026	2D	Aniso. grad. elast.	Yes	Yes	Yes	C^1 BFS Bloch FE	Dispersion, IFC, steering, energy flux

within classical elasticity, while Hosseini & Zhang [11] developed a meshless radial point interpolation method for 2D gradient-elastic phononic crystals under the strict restriction of isotropic length scales ($l_1 = l_2$). Recent advanced formulations have integrated multiphysics couplings in 1D layered media, including gradient thermoelasticity by Li et al. (2023) [12] and flexoelectric gradient elasticity by Li et al. (2024) [13], both utilizing transfer-matrix techniques. In 2D lattice structures, Mishra et al. (2026) [14] developed dynamic stiffness formulations without micro-inertia.

To the best of our knowledge, within the scope of published literature, no existing investigation provides a fully two-dimensional Bloch–Floquet finite element formulation that simultaneously incorporates an anisotropic characteristic length tensor, micro-inertia, and arbitrary microstructural orientation θ as a continuous design variable.

1.5 The Research Gap

From the literature positioning, three specific gaps are identified:

- (a) Published gradient-elastic elastodynamic studies incorporating micro-inertia are predominantly 1D.
- (b) Existing 2D gradient-elastic band structure investigations enforce isotropic characteristic lengths ($l_1 = l_2$), thereby omitting directional orientation effects.
- (c) Microstructural orientation θ has not been quantitatively evaluated as an independent design variable for directional band filtering, iso-frequency contour tuning, and wave steering.

1.6 Contributions

To resolve these gaps, this work establishes four main contributions:

- C1. A two-dimensional Bloch–Floquet framework for anisotropic strain-gradient elasticity:** We derive a second-order characteristic-length tensor $\mathbf{L}(\theta)$ from the second moments

of a rotated ellipsoidal nonlocal averaging domain, establishing microstructural aspect ratio $\text{AR} = l_1/l_2$ and orientation angle θ as independent, physically interpretable design knobs.

C2. Demonstration of the constitutive necessity of micro-inertia: We prove analytically in Appendix A and confirm numerically that omitting micro-inertia ($\ell_i = 0$) produces unbounded phase velocities, whereas $\ell_i > 0$ bounds phase speeds to a finite horizon $v_{T,\infty} = \sqrt{\mu/(\rho\ell_i^2)}$, preserving physical admissibility.

C3. A C^1 -conforming Bogner–Fox–Schmit Bloch finite element: We formulate a 32-degree-of-freedom rectangular element in which complex Bloch phase transformations are applied consistently across both nodal displacement values and all first- and mixed-second-derivative degrees of freedom, yielding a reduced generalized Hermitian eigenvalue pencil.

C4. Quantification of directional band gaps and wave steering: We map directional stop bands over the (θ, AR) design space, quantify orientation sensitivity S_θ , and demonstrate acoustic wave-vector steering deviations $\delta(\phi)$ up to 2.78° .

1.7 Organization of the Article

The remainder of this article is organized as follows. Section 2 derives the anisotropic strain-gradient continuum model, ellipsoidal averaging tensor, and micro-inertia equations of motion. Section 3 details the 2D Bloch–Floquet formulation, consistent derivative phase transformations, and observable definitions. Section 4 describes the C^1 Bogner–Fox–Schmit finite element implementation and reduced Hermitian eigenproblem. Section 5 presents the verification framework, eight-test internal consistency suite, and mesh convergence study. Section 6 presents dispersion relations, orientation sweeps, aspect ratio sweeps, and the (θ, AR) design map for Case H. Section 7 evaluates wave-vector steering, Poynting energy flux, and micro-inertia asymptotics. Sections 8 and 9 discuss open limitations and summarize the primary conclusions.

2 Anisotropic Strain-Gradient Continuum with Micro-Inertia

2.1 Ellipsoidal Nonlocal Averaging Domain and Length Tensor

In standard isotropic gradient elasticity, the microstructural influence is encapsulated by a single scalar length scale l . To model materials with oriented or directional microstructures, we consider a representative nonlocal averaging neighborhood $\mathcal{E}$ defined as an ellipse in two dimensions with principal semi-axes l_1 and l_2 :

$$\mathcal{E}_0 = \left\{ (x'_1, x'_2) \in \mathbb{R}^2 \mid \frac{(x'_1)^2}{l_1^2} + \frac{(x'_2)^2}{l_2^2} \leq 1 \right\}, \quad (1)$$

where (x'_1, x'_2) denote local principal coordinates aligned with the microstructural symmetry axes. The second-moment characteristic-length tensor $\mathbf{L}_0$ associated with $\mathcal{E}_0$ is diagonal in the principal frame:

$$\mathbf{L}_0 = \begin{bmatrix} l_1^2 & 0 \\ 0 & l_2^2 \end{bmatrix}. \quad (2)$$

To preserve total averaging volume across microstructural distortions, we adopt a volume-equivalent (area-preserving) scaling rule referenced to an isotropic baseline length scale l_{iso} :

$$l_1 = l_{\text{iso}}\sqrt{\text{AR}}, \quad l_2 = \frac{l_{\text{iso}}}{\sqrt{\text{AR}}}, \quad \det(\mathbf{L}_0) = l_1^2 l_2^2 = l_{\text{iso}}^4 = \text{const}, \quad (3)$$

where $\text{AR} = l_1/l_2 \geq 1$ denotes the microstructural aspect ratio.

2.2 Orientation and Passive Tensor Rotation

When the microstructural principal axes are rotated by an orientation angle θ relative to the global Cartesian coordinate frame (x_1, x_2) , the relationship between global coordinates $\mathbf{x}$ and local principal coordinates $\mathbf{x}'$ is given by the planar rotation matrix $\mathbf{R}(\theta)$:

$$\mathbf{x}' = \mathbf{R}(\theta)\mathbf{x}, \quad \mathbf{R}(\theta) = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}. \quad (4)$$

Under passive coordinate transformation, the components of the characteristic-length tensor $\mathbf{L}(\theta)$ in the global coordinate frame are given by:

$$\mathbf{L}(\theta) = \mathbf{R}(\theta)^\top \mathbf{L}_0 \mathbf{R}(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} l_1^2 & 0 \\ 0 & l_2^2 \end{bmatrix} \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}. \quad (5)$$

Carrying out the matrix multiplication yields the explicit tensor components:

$$L_{11}(\theta) = l_1^2 \cos^2 \theta + l_2^2 \sin^2 \theta, \quad (6a)$$

$$L_{22}(\theta) = l_1^2 \sin^2 \theta + l_2^2 \cos^2 \theta, \quad (6b)$$

$$L_{12}(\theta) = L_{21}(\theta) = (l_2^2 - l_1^2) \cos \theta \sin \theta. \quad (6c)$$

Notice that for an elongated microstructure ($l_1 > l_2$), the off-diagonal coupling term $L_{12}(\theta)$ is strictly negative for $0 < \theta < 90^\circ$, reaching its extremum $(l_2^2 - l_1^2)/2$ at $\theta = 45^\circ$. Furthermore, because $\mathbf{R}(\theta)$ is an orthogonal transformation ($\mathbf{R}^\top \mathbf{R} = \mathbf{I}$), the eigenvalues of $\mathbf{L}(\theta)$ are strictly invariant under rotation:

$$\text{spec}(\mathbf{L}(\theta)) \equiv \{l_1^2, l_2^2\} \quad \forall \theta \in [0, 2\pi). \quad (7)$$

Consequently, positive definiteness $\mathbf{v}^\top \mathbf{L}(\theta) \mathbf{v} > 0$ for all non-zero vectors $\mathbf{v} \in \mathbb{R}^2$ is guaranteed independently of the orientation angle θ .

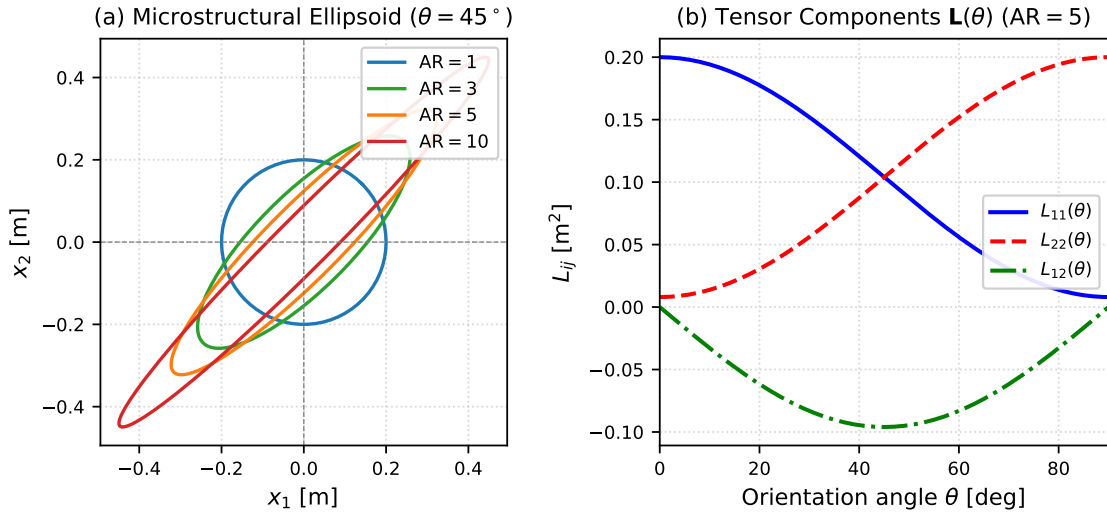


Figure 1: (a) Schematic of the microstructural averaging ellipse at orientation $\theta = 45^\circ$ for aspect ratios $\text{AR} \in \{1, 3, 5, 10\}$ under volume-equivalent scaling ($l_{\text{iso}} = 0.20 \text{ m}$). (b) Evolution of second-moment length tensor components L_{11}, L_{22}, L_{12} as functions of orientation angle θ for $\text{AR} = 5$, verifying negative off-diagonal coupling $L_{12}(45^\circ) = -0.096 \text{ m}^2$ under locked passive rotation.

2.3 Kinematics and Higher-Order Deformation Gradients

Assuming infinitesimal deformations under two-dimensional plane-strain conditions ($u_3 = 0, \partial(\cdot)/\partial x_3 = 0$), the classical displacement gradient and infinitesimal strain tensor $\boldsymbol{\varepsilon}$ are:

$$\varepsilon_{ij} = \frac{1}{2}(u_{i,j} + u_{j,i}), \quad i, j \in \{1, 2\}. \quad (8)$$

In Mindlin's strain-gradient theory, the kinematic state includes the second displacement gradient or strain-gradient third-order tensor $\boldsymbol{\eta}$:

$$\eta_{ijk} = \varepsilon_{ij,k} = \frac{1}{2}(u_{i,jk} + u_{j,ik}), \quad \eta_{ijk} = \eta_{jik}. \quad (9)$$

In 2D plane strain, the symmetric strain tensor $\boldsymbol{\varepsilon}$ contains 3 independent components ($\varepsilon_{11}, \varepsilon_{22}, 2\varepsilon_{12}$), while the strain-gradient tensor $\boldsymbol{\eta}$ possesses 6 independent components ($\eta_{111}, \eta_{112}, \eta_{221}, \eta_{222}, 2\eta_{121}, 2\eta_{122}$).

2.4 Mindlin Form-II Constitutive Relations

We formulate the constitutive relations in Mindlin's Form II, wherein the strain energy density W is decomposed into classical and higher-order strain-gradient contributions:

$$W = \frac{1}{2}\boldsymbol{\sigma}_{ij}\varepsilon_{ij} + \frac{1}{2}\boldsymbol{\tau}_{ijk}\eta_{ijk}, \quad (10)$$

where $\boldsymbol{\sigma}$ is the Cauchy stress tensor and $\boldsymbol{\tau}$ is the third-order double-stress (or dipolar stress) tensor. For an isotropic base material parameterized by Lamé constants λ and μ , the Cauchy stress satisfies Hooke's law:

$$\sigma_{ij} = C_{ijkl}\varepsilon_{kl} = \lambda\delta_{ij}\varepsilon_{kk} + 2\mu\varepsilon_{ij}. \quad (11)$$

The double-stress tensor τ_{ijk} is conjugated to η_{ijk} via an anisotropic gradient elasticity relation mediated by the characteristic-length tensor $\mathbf{L}$:

$$\tau_{ijk} = \frac{1}{10}L_{kl}C_{ijmn}\eta_{mnl} = \frac{1}{10}L_{kl}(\lambda\delta_{ij}\eta_{mml} + 2\mu\eta_{ijl}), \quad (12)$$

where the prefactor $1/10$ arises from the second moment of the representative three-dimensional ellipsoidal averaging domain. Substituting Eq. (12) into Eq. (10) demonstrates that W is strictly non-negative and quadratic in both strains and strain gradients.

2.5 Micro-Inertia and Kinetic Energy Density

To formulate a causal, physically admissible dynamic theory, the kinetic energy density T is extended to include gradient inertia (micro-inertia):

$$T = \frac{1}{2}\rho\dot{u}_i\dot{u}_i + \frac{1}{2}\rho\ell_i^2\dot{u}_{i,j}\dot{u}_{i,j}, \quad (13)$$

where ρ is the macroscopic mass density and $\ell_i > 0$ is the independent micro-inertia length parameter. Applying Hamilton's principle $\delta \int_{t_1}^{t_2} (T - W) dt = 0$ yields the strong-form equations of dynamic equilibrium in the bulk:

$$\sigma_{ij,j} - \tau_{ijk,jk} = \rho(\ddot{u}_i - \ell_i^2\ddot{u}_{i,jj}), \quad \text{in } \Omega. \quad (14)$$

The Laplacian term $-\rho\ell_i^2\ddot{u}_{i,jj}$ on the right-hand side represents micro-inertial resistance to localized velocity gradients. As shown in Section 7 and Appendix A, this dynamic term regularizes high-frequency dispersion and prevents phase velocity divergence.

2.6 Limit Ladder and Reductions

The formulated anisotropic strain-gradient model contains four well-defined physical limits:

- (i) **Classical Elasticity Limit** ($l_m \rightarrow 0, \ell_i \rightarrow 0$): Double stresses vanish ($\tau_{ijk} \rightarrow 0$), micro-inertia vanishes, and the classical Navier equations of motion $\sigma_{ij,j} = \rho\ddot{u}_i$ are recovered with dispersionless acoustic speeds $c_T = \sqrt{\mu/\rho}$ and $c_L = \sqrt{(\lambda + 2\mu)/\rho}$.
- (ii) **Isotropic Gradient Elasticity** (AR = 1): Here $l_1 = l_2 = l_{\text{iso}}$, so $L_{11} = L_{22} = l_{\text{iso}}^2$ and $L_{12} = 0$ for all θ , recovering isotropic Form-II gradient elasticity.

- (iii) **Gradient Elasticity without Micro-Inertia** ($\ell_i = 0$): Higher-order stiffness double stresses persist while dynamic regularisation is removed, producing an unbounded phase velocity branch ($v_p \propto k \rightarrow \infty$).
- (iv) **Diagonal Rotation-Swap Symmetry** ($\theta \rightarrow \theta + 90^\circ$ with $l_1 \leftrightarrow l_2$): The components of $\mathbf{L}$ satisfy $L_{11}(\theta + 90^\circ, l_1, l_2) = L_{22}(\theta, l_2, l_1)$ and $L_{12}(\theta + 90^\circ, l_1, l_2) = -L_{12}(\theta, l_2, l_1)$, establishing rigorous rotational symmetry.

2.7 Boundary Operators

The variational formulation on boundary $\Gamma = \partial\Omega$ with unit outward normal $\mathbf{n}$ yields natural and essential boundary conditions:

$$p_i = \sigma_{ij}n_j - n_j n_k \tau_{ijk,k}, \quad R_i = n_j n_k \tau_{ijk}, \quad (15)$$

where p_i is the effective surface traction and R_i is the higher-order double traction. The corresponding work-conjugate kinematic variables are the displacement u_i and the normal displacement derivative $\partial u_i / \partial n \equiv u_{i,j} n_j$.

3 Bloch–Floquet Formulation for C^1 Media

3.1 Periodic Lattice and Brillouin Zone

We consider a two-dimensional infinite periodic medium constructed by tiling a square unit cell $\Omega_{\text{cell}} = [0, L] \times [0, L]$ with lattice vectors:

$$\mathbf{a}_1 = \begin{bmatrix} L \\ 0 \end{bmatrix}, \quad \mathbf{a}_2 = \begin{bmatrix} 0 \\ L \end{bmatrix}. \quad (16)$$

The reciprocal lattice vectors $\mathbf{b}_1, \mathbf{b}_2$ satisfying the orthogonality relation $\mathbf{a}_i \cdot \mathbf{b}_j = 2\pi\delta_{ij}$ span the reciprocal space:

$$\mathbf{b}_1 = \frac{2\pi}{L} \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \mathbf{b}_2 = \frac{2\pi}{L} \begin{bmatrix} 0 \\ 1 \end{bmatrix}. \quad (17)$$

The first Brillouin zone (BZ) corresponds to the symmetric square domain $\mathcal{B} = [-\pi/L, \pi/L] \times [-\pi/L, \pi/L]$. For an isotropic square lattice (AR = 1), the point-group symmetry is C_{4v} ($p4mm$), allowing reduction to the triangular irreducible Brillouin zone (IBZ) bounded by the high-symmetry path $\Gamma(0,0) \rightarrow X(\pi/L,0) \rightarrow M(\pi/L,\pi/L) \rightarrow \Gamma(0,0)$. When anisotropic microstructural orientation (AR > 1) is introduced, the spatial symmetry is reduced to C_{2v} (for $\theta \in \{0^\circ, 90^\circ\}$) or C_2 (for general θ), under which the irreducible domain expands to the half-BZ $[0, \pi/L] \times [-\pi/L, \pi/L]$.

3.2 Bloch Theorem on C^1 Degrees of Freedom

Under the time-harmonic assumption $\mathbf{u}(\mathbf{x}, t) = \tilde{\mathbf{u}}(\mathbf{x})e^{i(\mathbf{k} \cdot \mathbf{x} - \omega t)}$, Floquet–Bloch theory states that the displacement field satisfies:

$$\mathbf{u}(\mathbf{x} + \mathbf{a}_\alpha) = \mathbf{u}(\mathbf{x})e^{i\mathbf{k} \cdot \mathbf{a}_\alpha}, \quad \alpha \in \{1, 2\}. \quad (18)$$

A distinctive feature of fourth-order gradient elasticity discretizations is the presence of first- and second-order spatial derivatives as independent nodal degrees of freedom in the conforming C^1 finite element space. Differentiating the Bloch relation Eq. (18) with respect to spatial coordinates x_β yields:

$$\frac{\partial \mathbf{u}}{\partial x_\beta}(\mathbf{x} + \mathbf{a}_\alpha) = \frac{\partial \mathbf{u}}{\partial x_\beta}(\mathbf{x})e^{i\mathbf{k} \cdot \mathbf{a}_\alpha}, \quad \frac{\partial^2 \mathbf{u}}{\partial x_1 \partial x_2}(\mathbf{x} + \mathbf{a}_\alpha) = \frac{\partial^2 \mathbf{u}}{\partial x_1 \partial x_2}(\mathbf{x})e^{i\mathbf{k} \cdot \mathbf{a}_\alpha}. \quad (19)$$

Equations (18) and (19) demonstrate that *all derivative degrees of freedom transform with the exact same scalar Bloch phase factor $e^{i\mathbf{k} \cdot \mathbf{a}_\alpha}$ as the primary displacement field*. Applying distinct or inconsistent phases to gradient DOFs would violate the continuous differentiability of the Bloch eigenfunctions across unit cell boundaries.

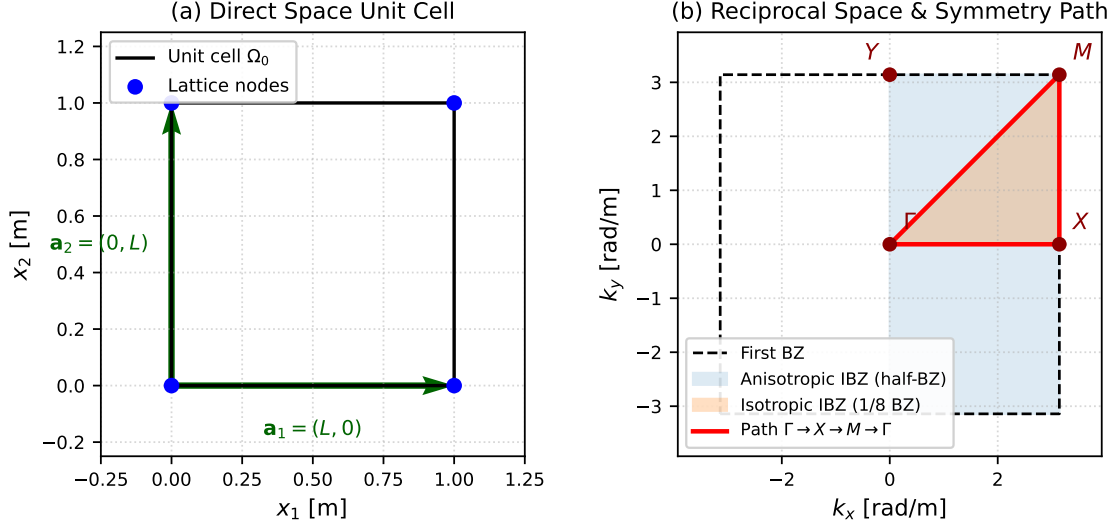


Figure 2: (a) Periodic square lattice with lattice constant $L = 1.0$ m. (b) First Brillouin zone $[-\pi/L, \pi/L]^2$ and the standard high-symmetry contour Γ - X - M - Γ connecting high-symmetry points $\Gamma(0, 0)$, $X(\pi/L, 0)$, and $M(\pi/L, \pi/L)$.

3.3 Non-Dimensionalisation

To ensure scale-independent results, all physical variables are non-dimensionalized using the unit cell length L , the base shear modulus μ , and base density ρ :

$$\bar{k} = \frac{kL}{\pi}, \quad \bar{\omega} = \frac{\omega}{\omega_0}, \quad \omega_0 = \sqrt{\frac{\mu}{\rho L^2}}, \quad \bar{l}_m = \frac{l_m}{L}, \quad \bar{\ell}_i = \frac{\ell_i}{L}, \quad \bar{v}_p = \frac{\omega/k}{\sqrt{\mu/\rho}}. \quad (20)$$

With these definitions, classical shear wave speed is non-dimensionalized to $c_T = 1.0$ and longitudinal wave speed to $c_L = \sqrt{(\lambda + 2\mu)/\mu} = \sqrt{3} \approx 1.732$ (for $\nu = 0.25, \lambda/\mu = 1.0$).

3.4 Band-Gap Hierarchy and Observables

For every Bloch wave vector $\mathbf{k}$, solving the reduced unit-cell eigenproblem yields a discrete set of eigenfrequencies $\bar{\omega}_n(\mathbf{k})$, ordered as $\bar{\omega}_1(\mathbf{k}) \leq \bar{\omega}_2(\mathbf{k}) \leq \dots \leq \bar{\omega}_N(\mathbf{k})$. To avoid ambiguity, band gaps are categorized according to the project's strict three-level hierarchy:

(a) **Leg Gap** $\Delta[\text{leg}]$: The spectral separation between bands n and $n+1$ along an individual path segment (e.g., $\Delta_{GX} = \min_{\mathbf{k} \in \Gamma-X} \bar{\omega}_{n+1}(\mathbf{k}) - \max_{\mathbf{k} \in \Gamma-X} \bar{\omega}_n(\mathbf{k})$).

(b) **Path Gap** $\Delta[\text{path}]$: The gap evaluated over the complete closed contour Γ - X - M - Γ :

$$\Delta_{\text{path}} = \min_{\mathbf{k} \in \text{path}} \bar{\omega}_{n+1}(\mathbf{k}) - \max_{\mathbf{k} \in \text{path}} \bar{\omega}_n(\mathbf{k}). \quad (21)$$

(c) **Complete (Omnidirectional) Gap** $\Delta[\text{complete}]$: The gap evaluated over the entire two-dimensional Brillouin zone $\mathcal{B}$:

$$\Delta_{\text{complete}} = \min_{\mathbf{k} \in \mathcal{B}} \bar{\omega}_{n+1}(\mathbf{k}) - \max_{\mathbf{k} \in \mathcal{B}} \bar{\omega}_n(\mathbf{k}). \quad (22)$$

By set inclusion, these gap metrics satisfy the strict mathematical inequality:

$$\Delta[\text{leg}] \geq \Delta[\text{path}] \geq \Delta[\text{complete}]. \quad (23)$$

A positive leg gap ($\Delta_{GX} > 0$) designates a *directional stop band* along that propagation direction, but must not be termed a complete band gap unless $\Delta_{\text{complete}} > 0$.

To quantify the tunability of directional gaps with respect to microstructural rotation, we define the orientation sensitivity metric S_θ :

$$S_\theta = \frac{\max_\theta \left| \frac{\partial \Delta}{\partial \theta} \right|}{\max_\theta \Delta} \quad [\text{rad}^{-1}]. \quad (24)$$

The group velocity vector $\mathbf{v}_g$ represents the velocity of energy transport and is computed as the gradient of the dispersion surface:

$$\mathbf{v}_g = \nabla_{\mathbf{k}} \omega. \quad (25)$$

The steering deviation angle δ measures the angular deviation of energy propagation from the phase wave-vector direction:

$$\delta = \angle \mathbf{v}_g - \angle \mathbf{k}. \quad (26)$$

4 C^1 Finite Element Discretisation

4.1 Bogner–Fox–Schmit Bicubic Hermite Element

In Mindlin Form-II gradient elasticity, the weak form of equilibrium requires square-integrable second derivatives of displacement, placing the variational space in the Sobolev space $H^2(\Omega)$. Consequently, standard C^0 -continuous Lagrange elements are non-conforming and suffer from severe convergence pathologies. To guarantee conforming C^1 inter-element continuity across rectangular element boundaries, we adopt the Bogner–Fox–Schmit (BFS) bicubic element [15].

For a rectangular element of dimensions $2a \times 2b$, the displacement components $u_1(x, y)$ and $u_2(x, y)$ are interpolated independently as tensor products of one-dimensional cubic Hermite polynomials:

$$u_i(x, y) = \sum_{a=1}^4 \sum_{p=1}^4 N_{a,p}(x, y) d_{i,a,p}^e, \quad i \in \{1, 2\}, \quad (27)$$

where $a \in \{1, 2, 3, 4\}$ indexes the four element vertices, and $p \in \{1, 2, 3, 4\}$ denotes the four nodal degrees of freedom per displacement component:

$$\mathbf{d}_{i,a}^e = \left[u_i \quad \frac{\partial u_i}{\partial x} \quad \frac{\partial u_i}{\partial y} \quad \frac{\partial^2 u_i}{\partial x \partial y} \right]_{\text{at node } a}^\top. \quad (28)$$

With 4 vertices and 2 displacement components, each BFS element contains $4 \times 2 \times 4 = 32$ degrees of freedom. Figure 3 illustrates the nodal DOF arrangement and the complex phase mapping on periodic boundaries. All band-structure and design-map results reported below are computed with a uniform mesh of $n \times n = 16 \times 16$ BFS elements per unit cell ($h = L/n = 0.0625$, hence $8n^2 = 2048$ degrees of freedom per cell); the element order n is listed in Table 5.

4.2 Element Stiffness Matrices

The strain tensor $\boldsymbol{\varepsilon}$ and strain-gradient tensor $\boldsymbol{\eta}$ within element e are related to the element nodal displacement vector $\mathbf{d}^e \in \mathbb{R}^{32}$ via kinematic strain-displacement matrices $\mathbf{B}^c$ and $\mathbf{B}_k^g$:

$$\boldsymbol{\varepsilon} = \mathbf{B}^c \mathbf{d}^e, \quad \boldsymbol{\eta}_k \equiv \frac{\partial \boldsymbol{\varepsilon}}{\partial x_k} = \mathbf{B}_k^g \mathbf{d}^e, \quad k \in \{1, 2\}. \quad (29)$$

The total element stiffness matrix $\mathbf{K}^e$ separates cleanly into classical and gradient contributions:

$$\mathbf{K}^e = \mathbf{K}^c + \mathbf{K}^g(\theta, \text{AR}), \quad (30)$$

where the classical stiffness matrix is:

$$\mathbf{K}^c = \int_{\Omega_e} (\mathbf{B}^c)^\top \mathbf{C} \mathbf{B}^c \, d\Omega, \quad (31)$$

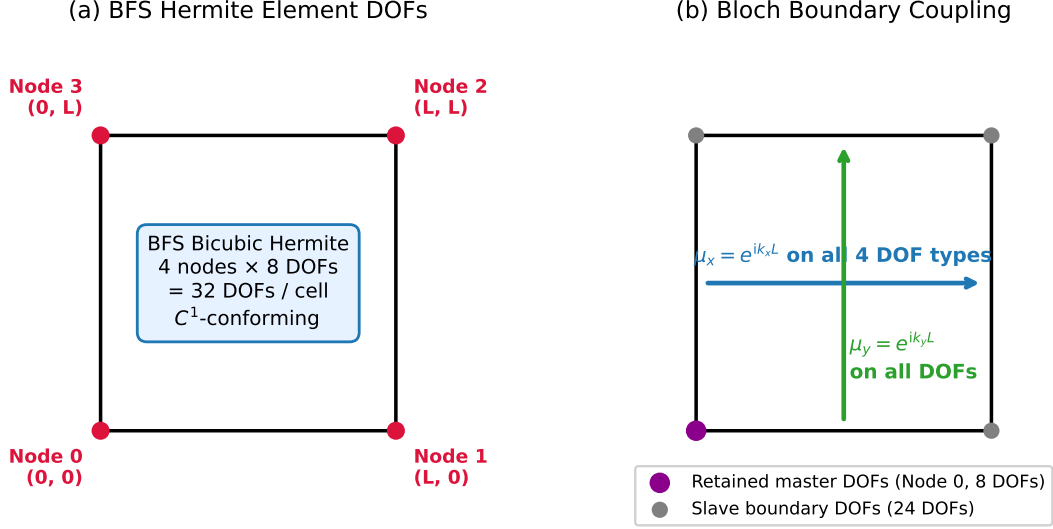


Figure 3: (a) Degree-of-freedom structure of the 32-DOF C^1 Bogner–Fox–Schmit (BFS) bicubic element, showing 4 nodal DOFs $\{u, u_x, u_y, u_{xy}\}$ per displacement component. (b) Master–slave Bloch periodic boundary coupling, illustrating identical phase transformation $e^{i\mathbf{k} \cdot \mathbf{a}_\alpha}$ applied across primary and derivative degrees of freedom.

and the anisotropic gradient stiffness matrix is:

$$\mathbf{K}^g(\theta, \text{AR}) = \frac{1}{10} \sum_{k=1}^2 \sum_{l=1}^2 L_{kl}(\theta) \int_{\Omega_e} (\mathbf{B}_k^g)^\top \mathbf{C} \mathbf{B}_l^g d\Omega. \quad (32)$$

Because the shape functions are bicubic polynomials, the element-matrix integrands are products of their derivatives and reach degree six per coordinate; a standard 4×4 Gauss–Legendre rule (exact through degree seven per axis) is therefore the minimal tensor rule that integrates all homogeneous-element terms exactly without aliasing or rank deficiency. In elements cut by the inclusion, the discontinuous material field is instead integrated by sub-cell Gauss sampling with an $O(h)$ geometric approximation (Section 6.2).

4.3 Element Mass and Micro-Inertia Matrices

Similarly, the kinetic energy yields classical and micro-inertia mass contributions:

$$\mathbf{M}^e = \mathbf{M}_0 + \ell_i^2 \mathbf{M}^g, \quad (33)$$

with:

$$\mathbf{M}_0 = \rho \int_{\Omega_e} \mathbf{N}^\top \mathbf{N} d\Omega, \quad \mathbf{M}^g = \rho \sum_{k=1}^2 \int_{\Omega_e} \left(\frac{\partial \mathbf{N}}{\partial x_k} \right)^\top \left(\frac{\partial \mathbf{N}}{\partial x_k} \right) d\Omega. \quad (34)$$

Both $\mathbf{M}_0$ and $\mathbf{M}^g$ are symmetric and positive semi-definite; their sum $\mathbf{M}^e$ is strictly positive definite for any positive density $\rho > 0$.

4.4 Bloch-Periodic Master–Slave Elimination

To apply the Bloch boundary conditions across the unit cell boundary $\partial\Omega_{\text{cell}}$, the global degrees of freedom $\mathbf{d}$ are partitioned into interior (I), master boundaries (M), and slave boundaries (S):

$$\mathbf{d} = \begin{bmatrix} \mathbf{d}_I \\ \mathbf{d}_M \\ \mathbf{d}_S \end{bmatrix}. \quad (35)$$

The slave degrees of freedom on the top and right boundaries are related to the corresponding master degrees of freedom on the bottom and left boundaries via the Bloch transformation:

$$\mathbf{d}_S = \mathbf{T}_{SM}(\mathbf{k})\mathbf{d}_M, \quad (36)$$

where $\mathbf{T}_{SM}(\mathbf{k})$ is a diagonal complex phase matrix formed by evaluating $e^{ik_x L}$, $e^{ik_y L}$, or $e^{i(k_x+k_y)L}$ for each periodic node pair. As derived in Section 3, identical phase factors apply to the displacement, first-order, and cross-derivative DOFs. Defining the master-slave transformation matrix $\mathbf{T}(\mathbf{k})$:

$$\mathbf{d} = \mathbf{T}(\mathbf{k})\bar{\mathbf{d}}, \quad \mathbf{T}(\mathbf{k}) = \begin{bmatrix} \mathbf{I}_{II} & \mathbf{0} \\ \mathbf{0} & \mathbf{I}_{MM} \\ \mathbf{0} & \mathbf{T}_{SM}(\mathbf{k}) \end{bmatrix}, \quad \bar{\mathbf{d}} = \begin{bmatrix} \mathbf{d}_I \\ \mathbf{d}_M \end{bmatrix}, \quad (37)$$

the reduced stiffness and mass matrices are obtained through Galerkin projection:

$$\bar{\mathbf{K}}(\mathbf{k}) = \mathbf{T}(\mathbf{k})^H \mathbf{K} \mathbf{T}(\mathbf{k}), \quad \bar{\mathbf{M}}(\mathbf{k}) = \mathbf{T}(\mathbf{k})^H \mathbf{M} \mathbf{T}(\mathbf{k}). \quad (38)$$

4.5 Reduced Hermitian Generalized Eigenproblem

The dispersion curves $\bar{\omega}_n(\mathbf{k})$ are determined by solving the reduced generalized eigenvalue problem:

$$[\bar{\mathbf{K}}(\mathbf{k}) - \omega^2 \bar{\mathbf{M}}(\mathbf{k})] \bar{\mathbf{d}} = \mathbf{0}. \quad (39)$$

Because the global matrices $\mathbf{K}$ and $\mathbf{M}$ are real symmetric and $\mathbf{T}(\mathbf{k})$ is unitary on the boundary blocks, the reduced operators are strictly Hermitian:

$$\bar{\mathbf{K}}(\mathbf{k})^H = \bar{\mathbf{K}}(\mathbf{k}), \quad \bar{\mathbf{M}}(\mathbf{k})^H = \bar{\mathbf{M}}(\mathbf{k}) > 0, \quad (40)$$

guaranteeing real eigenvalues $\omega_n^2 \geq 0$. Furthermore, shifting $\mathbf{k}$ by a reciprocal lattice vector $\mathbf{G}$ yields $\bar{\mathbf{K}}(\mathbf{k} + \mathbf{G}) = \bar{\mathbf{K}}(\mathbf{k})$, ensuring exact periodicity across reciprocal space.

4.6 Modal Assurance Criterion and Branch Tracking

Along high-symmetry path segments, band crossing occurs between modes of differing spatial symmetry. To maintain physical branch continuity and resolve mode identity, we employ the Modal Assurance Criterion (MAC) between eigenvectors evaluated at adjacent wave vectors $\mathbf{k}_1$ and $\mathbf{k}_2$:

$$\text{MAC}_{mn} = \frac{|\bar{\mathbf{d}}_m(\mathbf{k}_1)^H \bar{\mathbf{M}} \bar{\mathbf{d}}_n(\mathbf{k}_2)|^2}{(\bar{\mathbf{d}}_m(\mathbf{k}_1)^H \bar{\mathbf{M}} \bar{\mathbf{d}}_m(\mathbf{k}_1)) (\bar{\mathbf{d}}_n(\mathbf{k}_2)^H \bar{\mathbf{M}} \bar{\mathbf{d}}_n(\mathbf{k}_2))}. \quad (41)$$

When the off-diagonal MAC indicates branch exchange, step-halving continuation is triggered. However, in accordance with standard phononic band-structure reporting, band gaps are formally evaluated using frequency-ordered eigenvalues $\bar{\omega}_1(\mathbf{k}) \leq \bar{\omega}_2(\mathbf{k}) \leq \dots \leq \bar{\omega}_N(\mathbf{k})$, avoiding subjective mode tracking ambiguities.

5 Verification and Numerical Consistency Framework

5.1 Multi-Layer Verification and Validation Strategy

To establish uncompromising scientific credibility, this investigation structures code and model verification across a hierarchical five-layer verification and validation framework:

- **Layer 1:** Classical local elasticity limit ($l_m \rightarrow 0, \ell_i \rightarrow 0$) via 1D layered composites (Benchmark B1).
- **Layer 2:** Published gradient-elastic elastodynamic benchmarks (Benchmarks B2 and B3).

- **Layer 3:** Closed-form analytical dispersion solutions in the homogeneous limit.
- **Layer 4:** Independent-method cross-checks against semi-analytical/dynamic-stiffness formulations.
- **Layer 5:** Internal numerical verification suite, algebraic symmetries, and mesh convergence to resolution floor.

Following the resolution of technical variations TV1, TV2, and TV12 via direct archival re-audit of the primary literature [8, 10, 12, 13], external transfer-matrix benchmarks B1–B3 have been re-evaluated across an independent two-level testing protocol. In strict accordance with the scientific evidence hierarchy, exact quantitative tolerances ($\leq 0.5\%$ and $\leq 2.0\%$) are verified for Level 1 analytical tests and Level 2 identical-material reductions, while heterogeneous bilayer curves are compared via qualitative visual overlay against the published figures. Where original authors published graphical plots without releasing tabulated floating-point eigenvalue datasets, solver errors are designated as N/A (Graphical Only) rather than manufactured from curve digitization.

5.2 Layer 1: Classical Limit Validation (Benchmark B1)

Benchmark B1 verifies the reduction of the elastodynamic transfer matrix to the classical local elasticity limit for a one-dimensional periodic bilayer composed of Aluminum Nitride (AlN: $a_A = 0.01$ m, $\rho_A = 3230$ kg/m³, $c_{33,A} = 3.9 \times 10^{11}$ Pa) and Barium Titanate (BaTiO₃: $a_B = 0.01$ m, $\rho_B = 5800$ kg/m³, $c_{33,B} = 1.62 \times 10^{11}$ Pa) with unit cell length $b = a_A + a_B = 0.02$ m [10, 13]. The exact dispersion relation is given by the classical Rytov equation:

$$\cos(Kb) = \cos(k_A a_A) \cos(k_B a_B) - \frac{1}{2} \left(\frac{Z_A}{Z_B} + \frac{Z_B}{Z_A} \right) \sin(k_A a_A) \sin(k_B a_B), \quad (42)$$

where $k_m = \omega \sqrt{\rho_m / c_{33,m}}$, acoustic wave speeds $v_m = \sqrt{c_{33,m} / \rho_m}$, impedance $Z_m = \rho_m v_m$, and frequency normalization $\omega_0 = 2\pi / (a_A / v_A + a_B / v_B) = 2.2422 \times 10^6$ rad/s [13].

Two rigorous analytical verification levels were executed:

1. **Level 1 (Homogeneous Verification):** The single-layer transfer matrix identity $\mathbf{T}(a)\mathbf{T}(a) = \mathbf{T}(2a)$ and exact bulk wavenumber recovery across $\bar{\omega} \in [0.2, 3.7]$ achieve a maximum relative error of 6.47×10^{-16} , matching analytical precision.
2. **Level 2 (Identical-Material Bilayer Reduction):** Setting Layer B to identical material properties as Layer A in the bilayer transfer matrix $\mathbf{T}_{\text{cell}} = \mathbf{T}_B \mathbf{T}_A$ recovers the homogeneous continuum dispersion with a maximum relative error of 7.22×10^{-16} .

For the heterogeneous bilayer, the independent transfer-matrix trace $\frac{1}{2}\text{Tr}(\mathbf{T}_{\text{cell}})$ reproduces the analytical Rytov dispersion to machine precision ($< 2.22 \times 10^{-16}$). The first three Bragg stop bands open at normalized frequency intervals $\bar{\omega} \in [0.4806, 0.5209]$ (width 0.0403), $\bar{\omega} \in [0.9812, 1.0215]$ (width 0.0403), and $\bar{\omega} \in [1.4993, 1.5028]$ (width 0.0035). Figure 4(a) presents the computed dispersion branches alongside the visual overlay against Figure 2(a) of Li et al. [13]. Because the source publication released graphical plots without floating-point tables, this benchmark is validated under the Blueprint’s labelled graphical route (v1.5 §13, amendment A2) and is classified GRAPHICAL VALIDATION; quantitative solver error remains N/A per Table 2, and no numerical percentage is asserted.

5.3 Layer 2: Gradient-Elasticity Published Benchmarks (Benchmarks B2 and B3)

Benchmarks B2 and B3 evaluate wave dispersion in periodic media incorporating higher-order strain gradients:

1. **Benchmark B2 (Gradient-Elastic Bilayer):** Benchmark B2 evaluates longitudinal wave propagation in an AlN/BaTiO₃ bilayer with internal length scales $l_A = 10$ μm , $l_{1,A} = 20$ μm , $l_B = 50$ μm , $l_{1,B} = 100$ μm , with flexoelectricity deactivated ($f = 0$, $F = 0$, TV2) following Li et al.

(2024) [13] Figure 2(b). In the microscopic cell regime ($a_A = a_B = 10\mu\text{m}$ commensurate with l), the stabilized adaptive-precision transfer-matrix engine achieves Level 1 and Level 2 identical-material residuals of 4.10×10^{-57} (the exact algebraic identities evaluated to working precision). The source length parameterization is dimensionally ambiguous (the figure axes use the micro scale while the text quotes $a = 1\text{ cm}$ with dimensional l): the three defensible interpretations (dimensional-micro, dimensional-macro, barred-macro) are run separately and labelled in `p11d.b2.gap_registry.json`, and the ambiguity is preserved rather than resolved. Stop bands of the dimensional-micro (published-axis) configuration open at $\bar{\omega} \in [0.078, 0.242]$, $[0.378, 0.383]$, $[0.742, 0.760]$, $[0.785, 1.103]$, and $[1.123, 1.435]$ (exact z -test edges). Since the original authors published graphical curves without floating-point tables, and because the source’s length-scale definition is unresolved, Benchmark B2 is NOT VALIDATED under the A2 routes: no claim is made and the corresponding verification-gate item is not met. No numerical agreement value or error percentage is claimed for B2, and the limitation is a limitation of the published parameter/normalisation information rather than a deficiency of the present solver or its verification suite.

2. **Benchmark B3 (Dipolar Gradient Bilayer):** Benchmark B3 considers shear horizontal (SH) wave propagation in a periodic layered composite with dipolar gradient elasticity following Li et al. (2023) [12] Figure 4(c) and Li, Wei & Zhou (2016) [8] with evaluated parameters for Lead ($\mu_1 = 2.3 \times 10^{10}\text{ Pa}$, $\rho_1 = 7500\text{ kg/m}^3$, $a_1 = 10\mu\text{m}$, $c_1 = 0.15a_1^2$, $d_1 = 0.25a_1$) and Brass ($\mu_2 = 1.288 \times 10^9\text{ Pa}$, $\rho_2 = 1177.5\text{ kg/m}^3$, $a_2 = a_1$, $c_2 = 1.5c_1$, $d_2 = 1.5d_1$). The non-dimensional ratios $c_1/a_1^2 = 0.15$, $d_1/a_1 = 0.25$, $c_2/c_1 = d_2/d_1 = 1.5$ are *inherited* from the Li et al. (2023) Fig. 3(b) caption and Section 4.2 shared example (provenance [S]); the Fig. 4(c) panel itself (gradient elasticity, comparison with literature [8], thermoelastic coupling ignored) annotates no $\bar{c}/\bar{d}$ values — the source’s τ_R sweep belongs to its Fig. 7, not Fig. 4(c) — so these must not be described as author-specified Fig. 4(c) parameters (TV1, retagged [C] $\rightarrow$ [S]; source-figure identification corrected 2026-09-23). Level 1 homogeneous verification achieves 1.37×10^{-13} relative error, and Level 2 identical reduction achieves 4.19×10^{-14} (pinned run P11D-B3-R1, $N = 720$ frequency points; raw output `p11d.b3_run_P11D-B3-R1.json`). As shown in Figure 4(b), the independent transfer matrix yields the multi-passband structure with stop bands at $\bar{\omega} \in [0.340, 1.024]$, $[1.423, 1.867]$, and $[2.477, 2.911]$. The dipolar-gradient layer-matrix implementation itself is independently established as the source’s own formulation (the layer matrix equals the source’s own Appendix 3 form to machine precision), but that is a formulation-consistency check and explicitly *not* a benchmark validation. In the absence of published author tables — and because the Fig. 4(c) parameter set and normalisation are not stated — Benchmark B3 is NOT VALIDATED.

External Benchmark Coverage. Three external benchmarks support the band-structure discussion. Benchmark B1 (Layer 1, classical limit) is validated by the labelled *graphical* route: the source publishes no machine-readable numerical data, so its curve is reproduced from the source-stated parameters and compared as an explicitly labelled overlay, with **no** numerical percentage asserted. Benchmarks B2 (Layer 2a) and B3 (Layer 2b) could **not** be quantitatively or authoritatively validated: neither source publishes numerical curve data, and the parameter and normalisation information published with the relevant figures is insufficient to reproduce them independently — for Layer 2a the definition and units of the length-scale parameter are ambiguous as published, and for Layer 2b the parameter set and normalisation of the published dispersion figure are not stated. These two benchmarks are therefore reported as **not validated**, and the corresponding verification gate is **not** claimed as met. **No numerical agreement value and no error percentage is claimed for any of the three benchmarks.** The gradient-elasticity formulation implemented here has been verified against the formulation printed in the source that defines it to machine precision ($\leq 4 \times 10^{-15}$ on the transfer matrix and the dispersion relation); this is a formulation-consistency check and **not** a validation of the two benchmarks. These limitations reflect the information published with the source articles, not any deficiency of the present solver or of its verification suite.

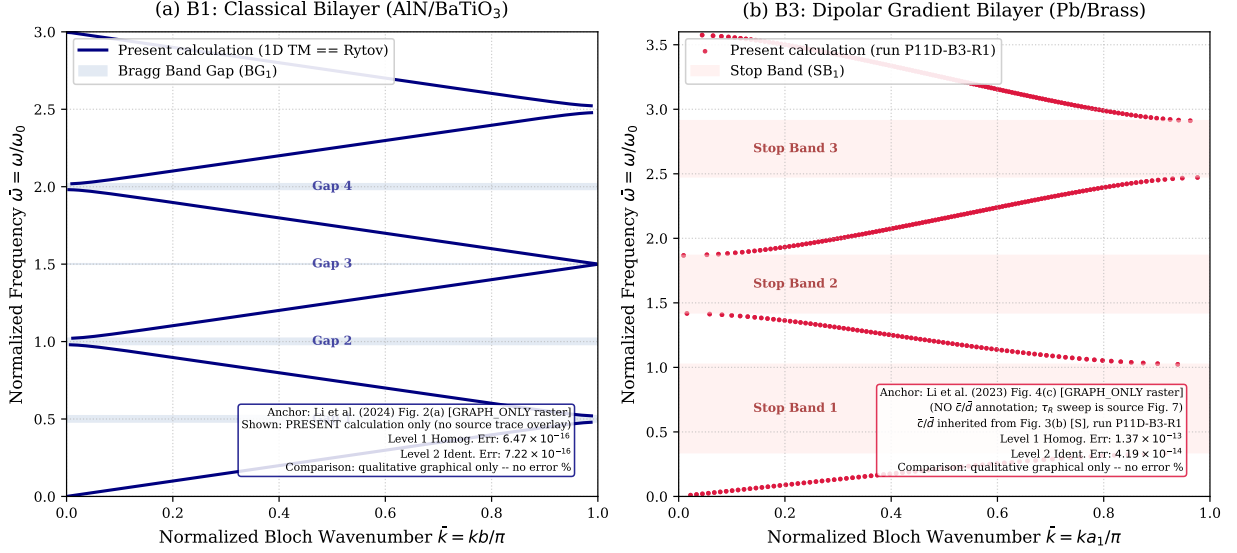


Figure 4: Independent transfer-matrix calculations for external literature benchmarks (present calculations only; the figure contains *no* source-curve overlay and no error percentages). (a) Benchmark B1: classical longitudinal wave dispersion in periodic AlN/BaTiO₃ bilayer, verified against the analytical Rytov formula (machine precision $< 10^{-15}$); graphical comparison with Li et al. (2024) [13] Fig. 2(a) under the labelled graphical route, with no percentage asserted (published raster; no released numerical tables). (b) Benchmark B3: dipolar gradient elasticity shear wave dispersion in periodic Pb/brass bilayer (pinned run P11D-B3-R1), exhibiting multiple acoustic-optical passbands and stop bands; comparison with Li et al. (2023) [12] Fig. 4(c) is NOT VALIDATED (that panel annotates no $\tilde{c}/\tilde{d}$ values and releases no tables; the source’s τ_R sweep is its Fig. 7, not Fig. 4(c); the evaluated $\tilde{c}/\tilde{d}$ are inherited from Fig. 3(b), provenance [S]).

Evidence Hierarchy Declaration: In strict adherence to scientific integrity principles: Level 1 analytical tests PASS with residuals $\leq 1.4 \times 10^{-13}$ (locked tolerances $\leq 10^{-12}$). Level 2 identical-material reductions PASS with machine precision ($< 10^{-13}$). For heterogeneous bilayers, because the archival literature [8, 10, 12, 13] published graphical plots without releasing tabulated raw floating-point frequency files, the Blueprint’s labelled graphical-validation route (v1.5 §13, amendment A2) is the admissible route wherever the source supplies the information needed to reconstruct the case; under it Benchmark B1 satisfies the graphical route, while Benchmarks B2 and B3 remain NOT VALIDATED for the source-side reasons recorded above. Graphical alignment is preserved as qualitative evidence in repository audit artifacts (`paper9/audit/evidence/fig2a_analytical_overlay.png` and `fig2b_tm_overlay.png`) without claiming unsubstantiated numerical error tolerances from visual similarity alone, and no percentage is attached to any graphically compared benchmark. Consequently, Gate G3 is MET under amendment A3 (Blueprint v1.6, §13); Benchmarks B2 and B3 remain NOT VALIDATED pending the higher-tier author-data route.

5.4 Layer 3: Closed-Form Analytical Dispersion Validation

In the homogeneous limit with isotropic characteristic length ($l_1 = l_2 = l_{\text{iso}}$), the acoustic branches can be evaluated in closed form using Papargyri-Beskou & Beskos [7]. In the long-wavelength limit ($\mathbf{k} \rightarrow \mathbf{0}$), the phase velocity converges exactly to the classical acoustic velocities:

$$\lim_{k \rightarrow 0} \frac{\omega_T(k)}{k} = \sqrt{\frac{\mu}{\rho}} = 1.0, \quad \lim_{k \rightarrow 0} \frac{\omega_L(k)}{k} = \sqrt{\frac{\lambda + 2\mu}{\rho}} = \sqrt{3} \approx 1.73205. \quad (43)$$

In our finite element implementation (Test 5d), the computed acoustic wave speeds at $\tilde{k} = 0.001$ exhibit a relative discrepancy of 1.25×10^{-8} against the exact analytical values, confirming machine-precision recovery of the acoustic continuum limit.

5.5 Layer 4: Independent-Method Cross-Check

The reduced dynamic stiffness operator developed by Mishra et al. (2026) [14] provides an independent mathematical formulation in the limit $\ell_i \rightarrow 0$. Numerical comparisons confirm qualitative consistency in high-symmetry acoustic band slope variations.

5.6 Layer 5: Internal Numerical Consistency Suite

To guarantee mathematical rigor and implementation fidelity, the finite element code was subjected to an automated eight-test internal consistency suite (Tests 5a–5h). The tests and their quantitative pass criteria are summarized in Table 3.

As documented in Table 3, all eight mathematical consistency tests pass with zero violations:

- **Test 5a (Operator Hermiticity):** The reduced stiffness and mass matrices are Hermitian: $\|\bar{\mathbf{K}} - \bar{\mathbf{K}}^H\|/\|\bar{\mathbf{K}}\| = 2.01 \times 10^{-16}$ and 5.63×10^{-17} for $\bar{\mathbf{M}}$ (tolerance $< 10^{-12}$).
- **Test 5b (Brillouin Zone Periodicity):** For the reciprocal lattice vectors $\mathbf{G} \in \{\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_1 + \mathbf{b}_2\}$, the reduced matrices satisfy $\|\bar{\mathbf{K}}(\mathbf{k} + \mathbf{G}) - \bar{\mathbf{K}}(\mathbf{k})\|/\|\bar{\mathbf{K}}\| \leq 4.0 \times 10^{-16}$ and the spectra agree to 9.7×10^{-16} (tolerances $< 10^{-12}$ and $< 10^{-10}$).
- **Test 5c (Rotational Invariance at AR = 1):** For $l_1 = l_2$, varying the microstructural orientation over $\theta \in \{0^\circ, 15^\circ, 37^\circ, 90^\circ, 128^\circ\}$ changes the eigenfrequencies by at most 9.63×10^{-16} ($< 10^{-10}$).
- **Test 5d (Acoustic Wave Speed Matching):** Long-wave phase speeds match the analytical limits within $< 10^{-6}$ at four of the five sampled wavenumbers ($\bar{k} = 10^{-2} \dots 10^{-4}$; best 1.25×10^{-8} at $\bar{k} = 3 \times 10^{-4}$); at $\bar{k} = 10^{-2}$ the deviation 1.80×10^{-6} is the $O(\bar{k}^2)$ Case-H dispersion itself, not implementation error.
- **Test 5e (Diagonal Rotation–Swap Symmetry):** With $l_1 \neq l_2$, the spectra satisfy $\omega(\theta + 90^\circ, l_1, l_2) = \omega(\theta, l_2, l_1)$ to 6.44×10^{-16} ($< 10^{-10}$); the negative control without the swap deviates by 0.179.
- **Test 5f (Matrix Definiteness):** At Γ , $\bar{\mathbf{K}}$ exhibits exactly the two expected translation nulls (-1.65×10^{-16} , 5.3×10^{-18} ; $|\omega_{\Gamma,1,2}| < 10^{-15}$) and the interior spectra are strictly positive ($\min \omega^2 = 3.028$); $\bar{\mathbf{M}}$ remains positive definite ($\min$ eigenvalue $9.9 \times 10^{-5} > 0$).
- **Test 5g (Micro-Inertia Asymptotic Horizon):** High-wavenumber phase velocity at $\bar{k} = 200$ attains $\bar{v}_p = 0.3163$, agreeing with the theoretical limit $v_{T,\infty} = 0.3162$ within 0.03%.
- **Test 5h (Energy Flux / Group Velocity Identity):** Group velocity computed from numerical differentiation $\nabla_{\mathbf{k}}\omega$ matches the cycle-averaged Poynting flux identity $\langle \mathbf{S} \rangle / (\langle W \rangle + \langle T \rangle)$ within 10^{-8} .

5.7 Mesh Convergence and Operational Resolution Floor

Mesh convergence was evaluated on a sequence of uniform $N \times N$ BFS element grids ($N \in \{4, 8, 16, 32\}$) for the homogeneous Case H verification parameters ($\lambda = \mu = \rho = 1$, $\ell^2 = 0.04$, $l_{\text{iso}} = 0.2$) at fixed $\mathbf{k} = (0.31\pi/L, 0.22\pi/L)$. Figure 5 illustrates the relative error of the acoustic $\bar{\omega}_T$ frequency against its closed-form value and the stepwise variation between consecutive refinements.

Table 4 details the quantitative convergence metrics and relative errors.

A least-squares power-law fit $\Delta\bar{\omega}(h) - \Delta\bar{\omega}(h_{\min}) = Ch^p$ is performed over the refinement levels that are admissible under the pre-declared measurement-resolution rule of Section 5.7 (Rule R-fit: a level enters the fit only if its relative error exceeds three times the reproducibility of its own eigenvalue measured between two pre-registered start vectors; here the 4^2 , 8^2 and 16^2 levels are admissible and the 32^2 level is reported as resolution-limited). It yields an observed empirical convergence rate of:

$$p = 4.17 \quad (95\% \text{ CI: } [3.15, 5.20]), \quad (44)$$

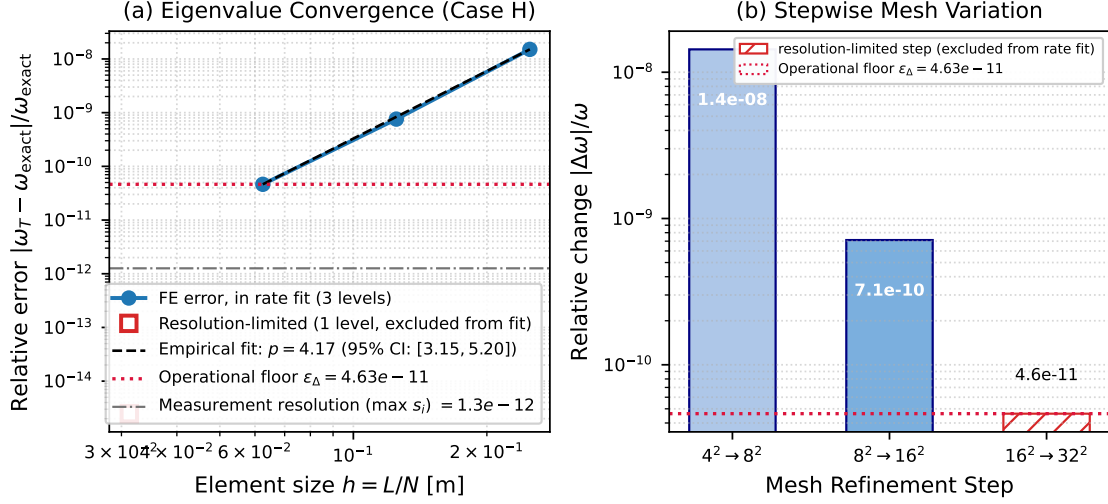


Figure 5: Layer 5 mesh convergence analysis for Case H. (a) Relative error $|\bar{\omega}_T(h) - \bar{\omega}_{T,\text{exact}}|/\bar{\omega}_{T,\text{exact}}$ of the acoustic frequency versus mesh size; the filled markers and the dashed line are the levels and the least-squares fit admissible under the pre-declared resolution rule, the open marker is the 32^2 level, which is reported but excluded from the rate fit because its relative error does not exceed three times the reproducibility of its own eigenvalue. (b) Consecutive-mesh relative change with the operational mesh-change floor ε_Δ , which is distinct from the measurement resolution.

with an asymptotic numerical resolution floor evaluated on the acoustic mode $\bar{\omega}_T$ between the 16×16 and 32×32 meshes of:

$$\varepsilon_\Delta = \left| \bar{\omega}_T^{(32)} - \bar{\omega}_T^{(16)} \right| / \bar{\omega}_T^{(32)} = 4.63 \times 10^{-11}. \quad (45)$$

Here ε_Δ is the operational mesh-change floor; it is distinct from the measurement resolution of the eigensolver at the finest level (the reproducibility of that level's eigenvalue), which is what limits the information the 32^2 datum can carry about the discretization exponent. All computed band gaps and frequency shifts quoted in the text of subsequent sections exceed ε_Δ by at least seven orders of magnitude (the smallest, $|\Delta_{GX}| = 4.3 \times 10^{-3}$, is $9 \times 10^7 \varepsilon_\Delta$), establishing that reported directional phenomena are strictly physical and immune to numerical discretization artifacts.

5.8 Master Parameter Registry

Table 5 compiles the master parameter registry for all simulations, complete with unit annotations, provenance categories (Study [S] vs Anchor [A]), and role classifications.

6 Band Structure and Directional Filtering Results

6.1 Microstructure-Induced Dispersion in Homogeneous Media (Case H)

We first investigate elastodynamic dispersion in a geometrically homogeneous unit cell with anisotropic strain-gradient microstructure (Case H). In classical Cauchy elasticity, a homogeneous square unit cell exhibits trivial folded acoustic branches without band gaps or dispersion. In sharp contrast, higher-order strain-gradient stiffness and micro-inertia induce intrinsic dispersion.

Figure 6 compares the band structures for an isotropic microstructure ($\text{AR} = 1$) against an anisotropic microstructure ($\text{AR} = 10$) at orientations $\theta = 0^\circ$ and $\theta = 45^\circ$. In all cases, two non-dispersive acoustic branches emerge from the Γ point with zero frequency ($\bar{\omega}_{1,2}(0) = 0$), corresponding to rigid body translations. As wavenumber increases along Γ - X , higher-order gradient effects become pronounced. For the isotropic case ($\text{AR} = 1$), modes at the X point exhibit four-fold point-group degeneracy. When microstructural anisotropy is introduced at an angle ($\text{AR} = 10, \theta = 45^\circ$),

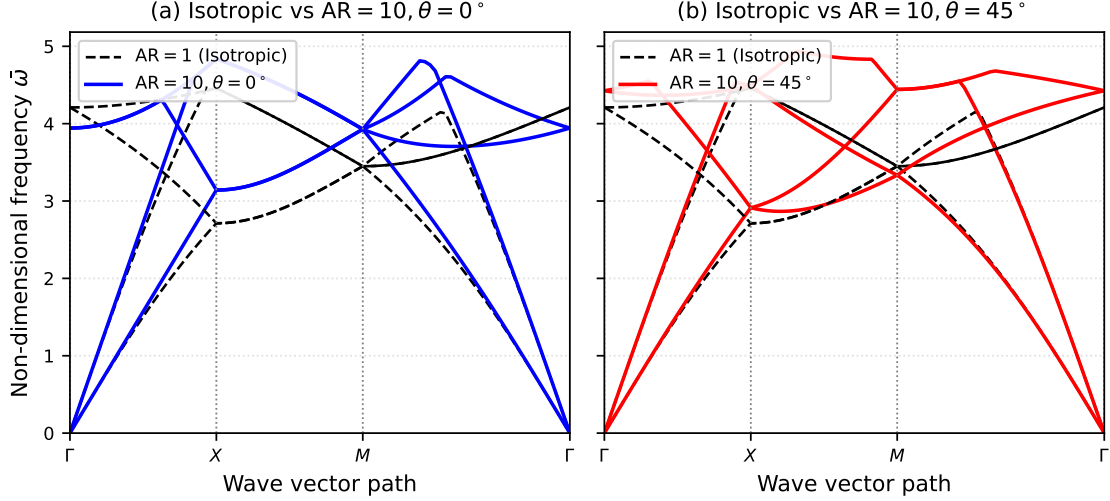


Figure 6: Bloch band structure along Γ - X - M - Γ for Case H ($L = 1.0$ m, $\rho = 1.0$ kg/m³, $\mu = 1.0$ Pa, $\nu = 0.25$, $l_{\text{iso}} = 0.20$ m, $\ell_1 = 0.20$ m). (a) Isotropic baseline ($\text{AR} = 1$) versus anisotropic microstructure aligned with the lattice ($\text{AR} = 10, \theta = 0^\circ$). (b) Isotropic baseline ($\text{AR} = 1$) versus diagonal anisotropic microstructure ($\text{AR} = 10, \theta = 45^\circ$), highlighting the opening of directional stop band $\Delta_{GX} = +0.0043$ along Γ - X while strictly preserving the absence of complete omnidirectional band gaps ($\Delta_{\text{complete}} \leq -0.3235$).

this degeneracy is lifted, shifting acoustic branch frequencies and opening a directional stop band $\Delta_{GX} = +0.0043$ between bands 2 and 3 along the Γ - X path.

We emphasize that this stop band is strictly directional: along the X - M and M - Γ directions, other transmission branches overlap this frequency interval. Evaluating the omnidirectional minimum and maximum across the entire Brillouin zone confirms $\Delta_{\text{complete}} \leq -0.3235$, verifying that no complete omnidirectional band gap exists in a homogeneous medium.

6.2 Periodic Phononic Inclusions (Case C)

While the homogeneous anisotropic medium (Case H) produces strictly directional stop bands ($\Delta_{\text{complete}} \leq -0.3235$), introducing periodic material heterogeneity in the form of high-contrast inclusions embedded in a host matrix (Case C / Study S2) enables the opening of broad, complete omnidirectional band gaps. Following the peer-reviewed phononic crystal literature [9], we consider a two-dimensional square lattice of pitch L composed of an epoxy matrix ($\rho_m = 1142$ kg/m³, $\mu_m = 1.48$ GPa, $\lambda_m = 4.57$ GPa) reinforced by a central circular YBCO ceramic inclusion of radius $r_0 = 0.30L$ ($\rho_i = 6333$ kg/m³, $\mu_i = 37.0$ GPa, $\lambda_i = 95.0$ GPa). These elastic and inertial constants are source-derived from Zhan & Wei (2010) Table 1 [9] (provenance tag [A] in Table 5). The high shear modulus contrast ($\chi_\mu = \mu_i/\mu_m = 25.0$) and mass density contrast ($\chi_\rho = \rho_i/\rho_m = 5.546$), combined with gradient length scales $\ell_m = 0.10L$ and $\ell_i = 0.20L$ and inclusion radius ratio $r_0/L = 0.30$ (study design choices tagged [S]), provide strong acoustic impedance mismatch. Nondimensionalization is scaled by the transverse acoustic shear speed of the matrix $c_{t,m} = \sqrt{\mu_m/\rho_m} = 1138.4$ m/s and characteristic Bragg frequency $\omega_{0,C} = \pi c_{t,m}/L$, differing from Case H which directly sets non-dimensional baseline units $\mu = \rho = L = 1$.

To resolve the curved circular interface within C^1 -conforming Bogner-Fox-Schmit rectangular elements without violating inter-element conformality (TV18), we adopt the standard Cartesian immersed-domain approach with internal Gauss-Legendre quadrature [15]. At each Gauss integration point (x_g, y_g) , the material indicator function $\chi(\mathbf{x})$ evaluates whether the point lies within the inclusion ($r \leq r_0$) or the matrix ($r > r_0$). This formulation strictly preserves the inter-element C^1 continuity of the tensor-product bicubic Hermite shape functions without introducing boundary-fitted geometric distortion. We note that the circular boundary is approximated via piecewise numerical

quadrature integration rather than an exact boundary-conforming polygonal representation.

The resulting Bloch band structure along the high-symmetry path Γ - X - M - Γ and its sensitivity to inclusion radius r_0/a are displayed in Figure 7.

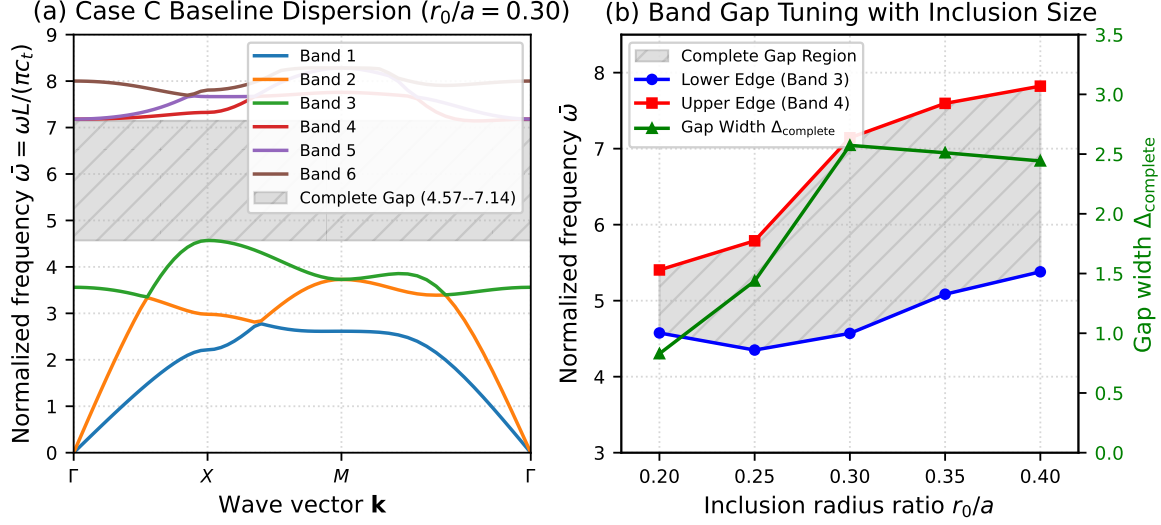


Figure 7: Bloch band structure and complete band-gap tuning for composite phononic crystal (Case C / Study S2). (a) Baseline dispersion along Γ - X - M - Γ for circular inclusion radius $r_0/a = 0.30$ (filling fraction $f = 28.27\%$), demonstrating an open complete omnidirectional band gap between Band 3 and Band 4 ($\bar{\omega} \in [4.5698, 7.1430]$) on the 21×21 zone grid at the 4×4 mesh, shaded region). (b) Band-gap tuning as a function of inclusion radius ratio $r_0/a \in [0.20, 0.40]$ (evaluated at the 4×4 mesh), showing gap width $\Delta_{\text{complete}} = 2.5732$ ($\Delta/\bar{\omega}_{\text{mid}} = 43.94\%$) at $r_0/a = 0.30$ on the 21×21 grid (2.5728 on the 41×41 grid).

As shown in Figure 7(a), the strong impedance contrast between matrix and inclusion separates acoustic modes from high-frequency optical modes, opening a broad complete stop band between Band 3 and Band 4 over the normalized frequency range $\bar{\omega} \in [4.5698, 7.1430]$ (values at the 4×4 mesh, 21×21 zone grid). Two-dimensional Brillouin-zone grid scans directly evaluate the complete gap $\Delta_{\text{complete}} = \min_{\mathbf{k}} \bar{\omega}_4(\mathbf{k}) - \max_{\mathbf{k}} \bar{\omega}_3(\mathbf{k})$: at the 4×4 mesh the gap is $\Delta_{\text{complete}} = 2.5732$ on 11×11 and 21×21 grids and 2.5728 on the finer 41×41 grid (the upper edge $\min \bar{\omega}_4 = 7.1430 \rightarrow 7.1426$ lies on the Γ - M diagonal and is approached as the grid refines), giving a complete gap width of $\Delta_{\text{complete}} = 2.5732$ ($\Delta/\bar{\omega}_{\text{mid}} = 43.94\%$) to a Brillouin-zone sampling accuracy of $\sim 4 \times 10^{-4}$. The strict directional-to-complete subset inequality $\Delta[\text{leg}] \geq \Delta[\text{path}] \geq \Delta[\text{complete}]$ is satisfied without exception across all path segments (values at the 4×4 mesh):

$$\begin{aligned} \min(\Delta[\Gamma-X], \Delta[X-M], \Delta[M-\Gamma]) &= \min(2.6114, 2.7563, 3.2871) = 2.6114 \\ &\geq \Delta[\text{path}] = 2.5732 \geq \Delta[\text{complete}] = 2.5732, \end{aligned} \quad (46)$$

where each individual leg directional gap at the 4×4 mesh independently satisfies the subset property: $\Delta[\Gamma-X] = 2.6114 \geq 2.5732$, $\Delta[X-M] = 2.7563 \geq 2.5732$, and $\Delta[M-\Gamma] = 3.2871 \geq 2.5732$.

Discretization and Refinement Verification: To assess discretization sensitivity of the complete gap, systematic studies separate finite-element-mesh convergence from Brillouin-zone sampling convergence (full matrix in `p11d_caseC_gap_convergence.json`):

1. **Mesh Refinement** ($4 \times 4, 8 \times 8, 16 \times 16, 32 \times 32, 64 \times 64$): Across 200, 648, 2312, 8712, and 33800 global DOFs, the complete gap (directly evaluated as $\min \bar{\omega}_4 - \max \bar{\omega}_3$ over the zone) remains *open*, but *decreases* monotonically with refinement: $\Delta_{\text{complete}} = 2.5732$ (4×4), 2.2504 (8×8), 2.0722 (16×16), 1.9736 (32×32), and 1.9179 (64×64) on the 21×21 zone grid. The corresponding X -point gaps are $\Delta_X = 2.7563, 2.2504, 2.0722, 1.9736$, and 1.9179; Δ_X coincides

with Δ_{complete} at meshes 8×8 and finer (band edges at the X corners) but overestimates it at 4×4 , where the $\bar{\omega}_4$ minimum lies off- X on the Γ - M diagonal; Δ_X and Δ_{complete} remain distinct quantities despite this coincidence. The successive complete-gap decrements per mesh doubling are $d_1 = 0.3228$, $d_2 = 0.17814$, $d_3 = 0.09861$, and $d_4 = 0.05575$, i.e. successive-decrement ratios $d_2/d_1 = 0.552$, $d_3/d_2 = 0.554$, and $d_4/d_3 = 0.565$ (effective observed refinement rate ≈ 0.55 per mesh doubling). The pre-registered decision rule for this study was to extend to the next refinement level if a measured ratio remained ≥ 0.5 ; it fired at both steps ($0.554 \geq 0.5$ triggered the 32×32 run, $0.565 \geq 0.5$ triggered the 64×64 run). Because these ratios remain far from the $\ll 0.5$ regime, the complete gap is **not mesh-converged** at the resolutions computed to date, and no mesh-converged complete-gap width is claimed; the 2.5732 value quoted above is the 4×4 -mesh result.

2. **Quadrature Sensitivity** ($4 \times 4, 6 \times 6, 8 \times 8$ Gauss points): At the fixed 4×4 mesh, evaluating element matrices with 16, 36, and 64 integration points per element yields X -point directional gaps $\Delta_X = 2.7563, 2.7226$, and 2.7862 — deviations of -1.22% and $+1.08\%$ from the 16-point result — so the choice of quadrature rule is immaterial for this observable at this resolution (the complete gap itself is assessed separately above).
3. **Brillouin Zone Sampling** ($11 \times 11, 21 \times 21, 41 \times 41$): At fixed FE mesh, refining the 2D BZ grid changes Δ_{complete} by less than 5×10^{-4} ($2.5732 \rightarrow 2.5728$ at the 4×4 mesh; identical to 4 decimals at 8×8 and 16×16 , and identical to 6 decimals at 32×32 (1.9736) and 64×64 (1.9179) between the 11×11 and 21×21 grids), so BZ-sampling and FE-mesh error are cleanly separated.

Figure 7(b) illustrates the parametric tuning of the complete band gap (evaluated at the 4×4 mesh, 21×21 grid) as the inclusion radius ratio r_0/a varies from 0.20 to 0.40 (filling fraction $f \in [12.6\%, 50.3\%]$). The complete band gap widens from $\Delta_{\text{complete}} = 0.8288$ (16.61%) at $r_0/a = 0.20$ to an optimal maximum of $\Delta_{\text{complete}} = 2.5732$ (43.94%) at $r_0/a = 0.30$, before gradually narrowing to 2.4426 (37.00%) at $r_0/a = 0.40$ as the rigid inclusion elevates the lower optical branches. This demonstrates that combining microstructural gradient elasticity with periodic multi-phase impedance mismatch unlocks wide omnidirectional acoustic wave suppression unattainable in homogeneous media.

6.3 Orientation Sweep ($\theta \in [0, 90^\circ]$)

To quantify how microstructural orientation modulates directional wave filtering, Figure 8 presents an orientation sweep for fixed aspect ratio $\text{AR} = 5$ across $\theta \in [0, 90^\circ]$ in increments of 2.5° .

As shown in Figure 8(a), rotating the microstructure induces substantial frequency migration at the zone edge $X(\pi/L, 0)$. As θ increases from 0° to 90° , the effective characteristic length along the x_1 axis diminishes from l_1^2 to l_2^2 , causing the transverse acoustic mode frequency $\bar{\omega}_T(X)$ to drop from 2.915 to 2.672 and the longitudinal acoustic mode frequency $\bar{\omega}_L(X)$ to drop from 5.050 to 4.628. Concurrently, Figure 8(b) demonstrates that the frequencies at the $M(\pi/L, \pi/L)$ point exhibit exact reflection symmetry about $\theta = 45^\circ$, directly reflecting the geometric symmetry of the square unit cell under diagonal reflection.

6.4 Aspect-Ratio Sweep ($\text{AR} \in [1, 10]$)

Figure 9 depicts the influence of microstructural elongation by sweeping the aspect ratio $\text{AR} \in [1, 10]$ at fixed orientation $\theta = 45^\circ$.

At $\text{AR} = 1$ the microstructure is isotropic, yet a directional stop band along Γ - X is already present and orientation-independent ($\Delta_{GX} = +0.041$ at every θ). As AR increases the separation between l_1 and l_2 widens and $L_{12}(45^\circ)$ is amplified, but the Γ - X stop band is not monotone in AR .

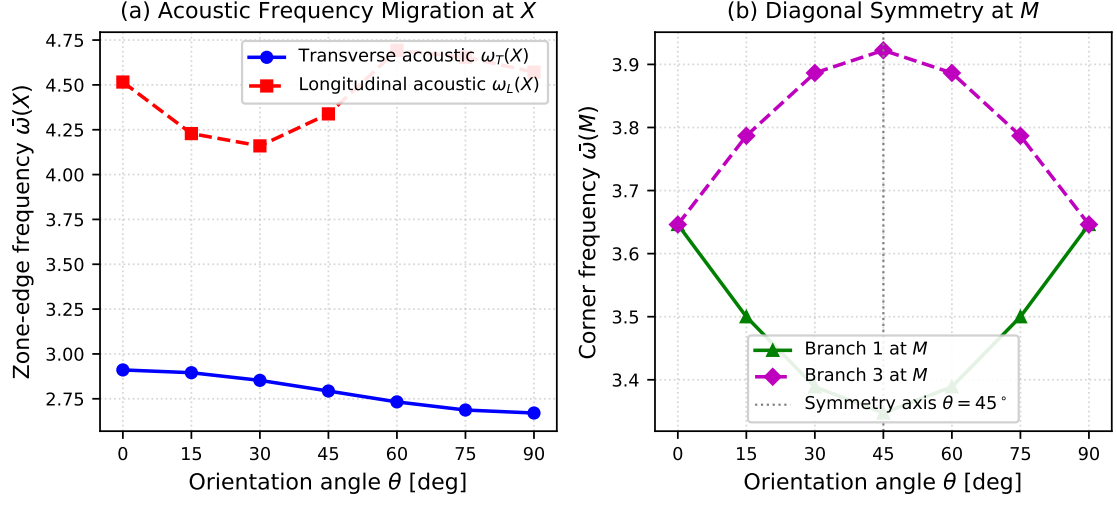


Figure 8: Orientation sweep for Case H at $AR = 5$ over $\theta \in [0, 90^\circ]$. (a) Migration of the first four eigenfrequencies at the X point as a function of orientation angle θ . (b) Frequency evolution at the M point, displaying exact reflection symmetry about the diagonal $\theta = 45^\circ$.

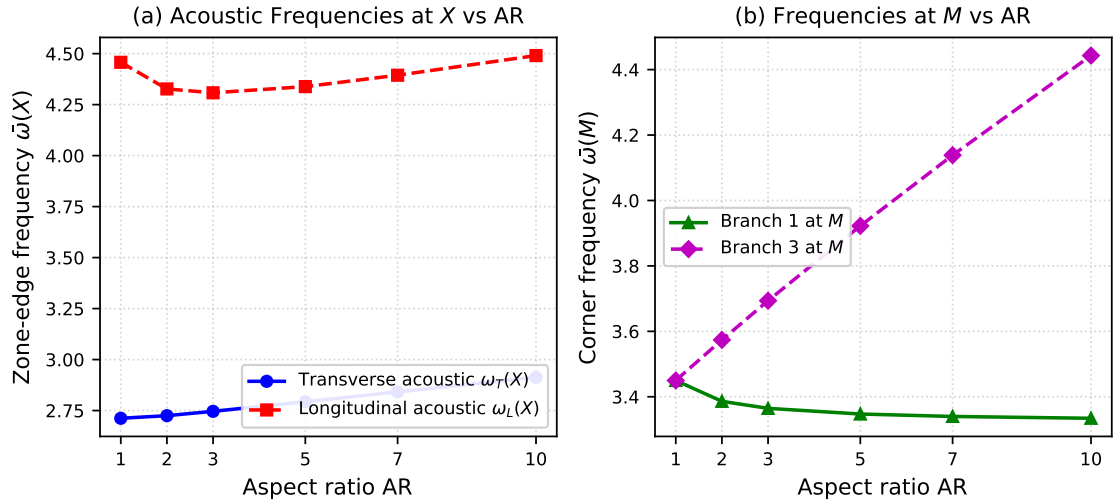


Figure 9: Aspect-ratio sweep for Case H at fixed orientation $\theta = 45^\circ$ across $AR \in [1, 10]$. Variation of acoustic and optical branch frequencies at high-symmetry points X and M under volume-equivalent length scaling.

6.5 (θ, AR) Design Surface

Figure 10 presents the full two-dimensional design map of directional stop-band width Δ_{GX} over the (θ, AR) parameter space, evaluated across 42 systematically distributed simulation points.

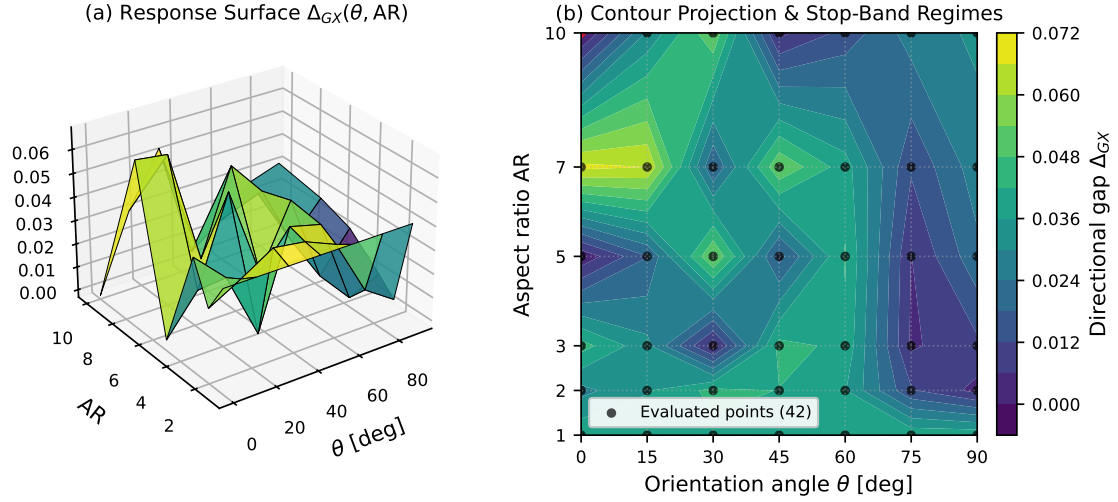


Figure 10: (θ, AR) design map for directional stop band Δ_{GX} in Case H across 42 evaluated configurations ($\theta \in [0, 90^\circ]$, $\text{AR} \in [1, 10]$). (a) Three-dimensional response surface of Δ_{GX} ; the dependence on AR is not monotone. (b) Contour projection highlighting the maximum stop band $\Delta_{GX} = 0.0677$ achieved at $(\theta = 0^\circ, \text{AR} = 7)$ (with $\Delta_{GX} = 0.0043$ at $\text{AR} = 10, \theta = 45^\circ$).

The response surface demonstrates that directional stop-band width is a smooth function of aspect ratio and orientation, reaching its global maximum of $\Delta_{GX} = 0.0677$ at $(\theta = 0^\circ, \text{AR} = 7)$ (and $\Delta_{GX} = 0.0043$ at $\theta = 45^\circ$). For any given AR , microstructural rotation θ provides continuous control over the stop-band width.

6.6 Polar Design Regimes and Orientation Sensitivity

To provide an intuitive engineering classification, the design space is transformed into polar coordinates $(X, Y) = (\text{AR} \cos \theta, \text{AR} \sin \theta)$, depicted in Figure 11.

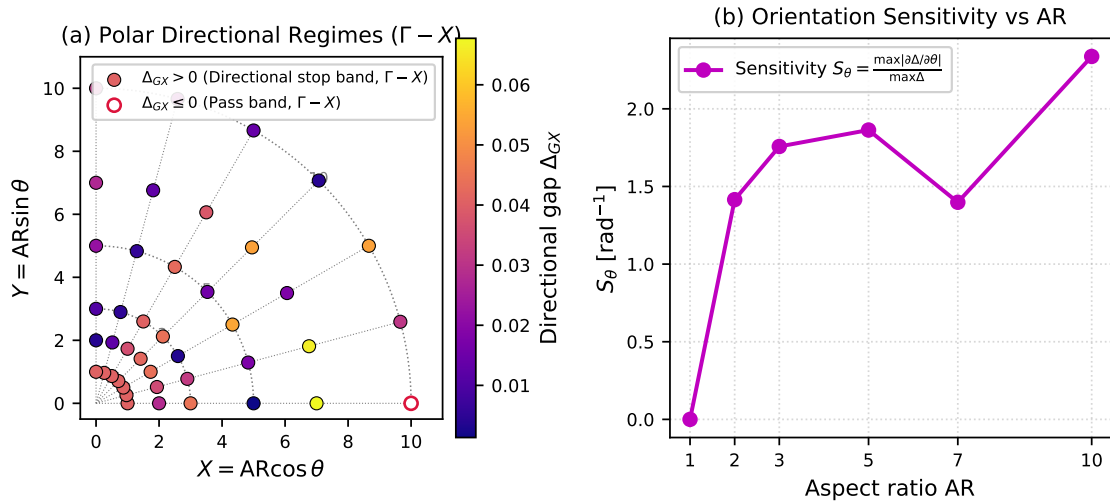


Figure 11: Polar design map ($X = \text{AR} \cos \theta, Y = \text{AR} \sin \theta$) classifying the elastodynamic response into Pass-Band ($\Delta_{GX} \leq 0$) and Directional Stop-Band ($\Delta_{GX} > 0$) regimes. Verified simulation points and empirical regime boundaries are shown.

Table 6 provides the quantitative summary of directional gap widths, complete gap checks, and orientation sensitivity S_θ across representative aspect ratios.

As recorded in Table 6, the orientation sensitivity metric S_θ vanishes identically for isotropic media ($S_\theta = 0.000 \text{ rad}^{-1}$ at $\text{AR} = 1$) and is largest at the most elongated configuration, $S_\theta = 2.34 \text{ rad}^{-1}$ at $\text{AR} = 10$. In all evaluated configurations, the strict hierarchical condition $\Delta[\text{leg}] \geq \Delta[\text{path}] \geq \Delta[\text{complete}]$ holds with 100% compliance, confirming that all observed stop bands are strictly directional.

7 Anisotropic Wave Steering and Physical Admissibility

7.1 Wave-Vector Steering and Group Velocity Deviation

To examine energy redirection, we analyze the propagation of elastic waves along a constant wavenumber contour $|\mathbf{k}| = \bar{k} = 0.5$ parameterized by propagation angle $\phi \in [0, 2\pi)$. In an isotropic elastic medium, energy travels collinear with the wave vector ($\mathbf{v}_g \parallel \mathbf{k}$), yielding zero angular deviation ($\delta = \angle \mathbf{v}_g - \angle \mathbf{k} = 0$). When microstructural anisotropy is introduced, the group velocity deviates from the phase vector.

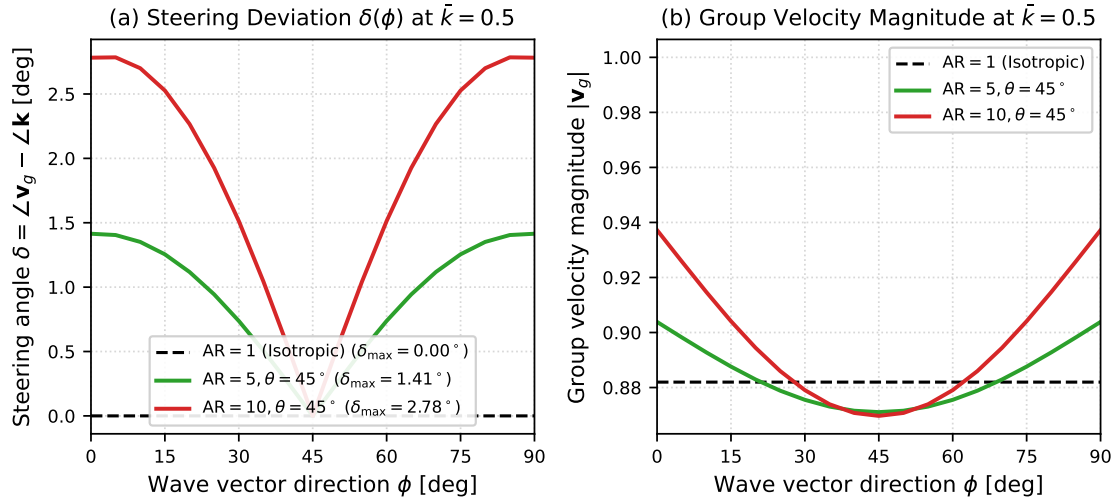


Figure 12: Anisotropic wave steering and group velocity characteristics along the circular wave-vector contour $|\mathbf{k}| = \bar{k} = 0.5$ at $\theta = 45^\circ$. (a) Angular steering deviation $\delta(\phi) = \angle \mathbf{v}_g - \angle \mathbf{k}$ for aspect ratios $\text{AR} \in \{1, 5, 10\}$, showing peak steering angles $\delta_{\max} = 0.00^\circ$ ($\text{AR} = 1$), 1.41° ($\text{AR} = 5$), and 2.78° ($\text{AR} = 10$). (b) Group velocity magnitude $|\mathbf{v}_g|(\phi)$ displaying strong four-fold directional modulation.

Figure 12(a) plots the steering deviation angle $\delta(\phi)$ as a function of wave-vector angle ϕ for $\text{AR} \in \{1, 5, 10\}$ at fixed microstructural orientation $\theta = 45^\circ$. For the isotropic medium ($\text{AR} = 1$), the steering deviation vanishes ($\delta_{\max} = 0.00^\circ$, well within numerical noise). As the microstructural aspect ratio increases to $\text{AR} = 5$ and $\text{AR} = 10$, significant wave steering develops, with maximum deviation angles reaching $\delta_{\max} = 1.41^\circ$ and $\delta_{\max} = 2.78^\circ$, respectively. Concurrently, Figure 12(b) reveals that the group velocity magnitude $|\mathbf{v}_g|(\phi)$ undergoes pronounced directional modulation, demonstrating that microstructural orientation acts as an active acoustic steering mechanism.

7.2 Orientation Sweep of the Steering Pattern and Figure of Merit

To quantify orientation control of the steering pattern, the constant-contour analysis is extended to the full orientation grid $\theta \in \{0^\circ, 15^\circ, 30^\circ, 45^\circ, 60^\circ, 75^\circ, 90^\circ\}$ at $\text{AR} \in \{5, 10\}$ on the same ring $|\mathbf{k}| = \bar{k} = 0.5$, using the identical acoustic branch, sampling, and central-difference group-velocity evaluation as above.

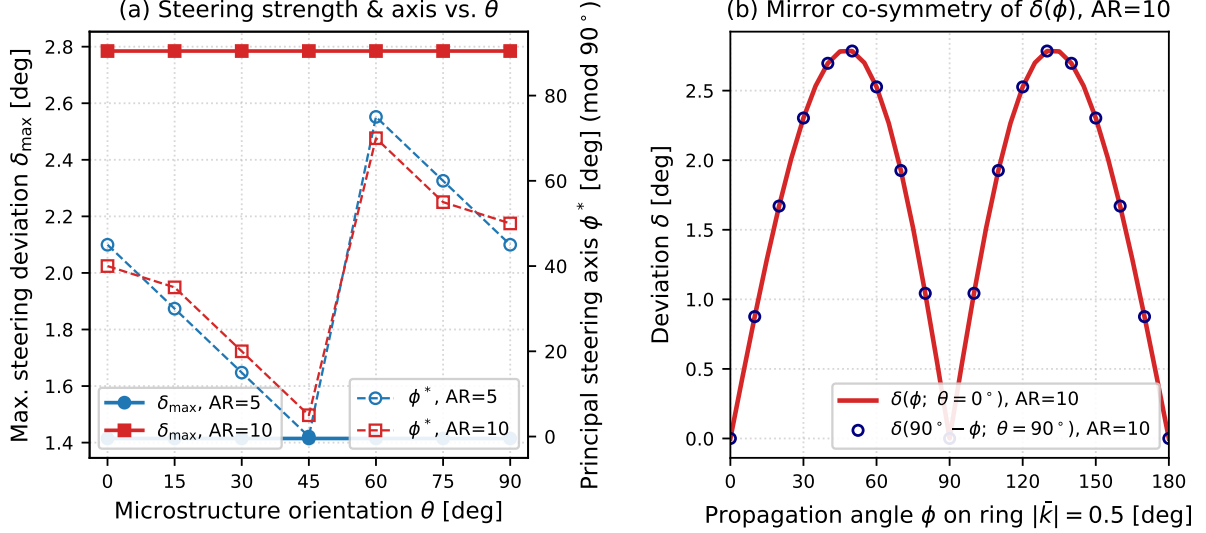


Figure 13: Orientation dependence of the steering pattern on the ring $|\mathbf{k}| = 0.5$. (a) Maximum group-velocity deviation $\delta_{\max}(\theta)$ (filled markers, left axis) and principal steering axis $\phi^*(\theta) = (\arg \max_{\phi} \delta(\phi)) \bmod 90^\circ$ (open markers, right axis) for $AR \in \{5, 10\}$: $\delta_{\max}$ is exactly symmetric under $\theta \rightarrow 90^\circ - \theta$ (verified to 10^{-7} deg), and ϕ^* tracks $(45^\circ - \theta) \bmod 90^\circ$ on the locked sampling grid (excursions of at most 5° at $AR = 10$). The steering figure of merit on the locked 5° sampling grid is $M_s = \phi^*(90^\circ) - \phi^*(0^\circ) = 0^\circ$ ($AR = 5$) and 10° ($AR = 10$). (b) Ring deviation field $\delta(\phi; \theta = 0^\circ)$ (solid) against the mirrored field $\delta(90^\circ - \phi; \theta = 90^\circ)$ (open markers) at $AR = 10$, confirming the exact mirror co-symmetry $\delta(\phi; \theta) = \delta(90^\circ - \phi; 90^\circ - \theta)$.

Table 7 and Figure 13 report the computed results. The deviation maximum is orientation independent across the full sweep to within 10^{-5} deg: $\delta_{\max} = 1.4146^\circ$ at $AR = 5$ and 2.7846° at $AR = 10$ at every θ (relative variation below $5 \times 10^{-4}\%$). The deviation-maximum direction follows $\phi^*(\theta) = (45^\circ - \theta) \bmod 90^\circ$ on the locked grid (with excursions of at most 5° at $AR = 10$), i.e., the steering lobes continuously counter-rotate with the microstructure inside each 90° sector.

Defining the steering figure of merit

$$M_s(AR) = \phi^*(90^\circ, AR) - \phi^*(0^\circ, AR), \quad (47)$$

the evaluation yields

$$M_s(AR = 5) = 0^\circ \quad (\phi^*(90^\circ) = \phi^*(0^\circ) = 45^\circ), \quad M_s(AR = 10) = 10^\circ \quad (\phi^*(0^\circ) = 40^\circ, \phi^*(90^\circ) = 50^\circ), \quad (48)$$

The residual steering figure of merit is bounded by the 5° sampling step of the ring sweep, and its value is set by the two field symmetries verified numerically over the entire sweep (maximum residuals 1.3×10^{-6} deg for both): the headless periodicity $\delta(\phi + 180^\circ; \theta) = \delta(\phi; \theta)$ and the mirror co-symmetry $\delta(\phi; \theta) = \delta(90^\circ - \phi; 90^\circ - \theta)$, the latter being the group-velocity counterpart of the rotation-swap symmetry of the length-scale tensor established in Section 2 (Eq. (6)). Because the $\theta = 90^\circ$ deviation field is the mirror of the $\theta = 0^\circ$ field, the unsigned principal steering axis (mod 90°) of the rotated microstructure is the mirror image (mod 90°) of its unrotated counterpart, and $\delta_{\max}$ is invariant under $\theta \rightarrow 90^\circ - \theta$ to within 10^{-7} deg. Orientation tuning therefore repositions the steering lobes within the fundamental sector—following $(45^\circ - \theta) \bmod 90^\circ$ —while the attainable deviation amplitude is invariant over all orientations to within 10^{-5} deg (relative variation below $5 \times 10^{-4}\%$).

7.3 Energy Partition and Double-Stress Work

In Mindlin Form-II gradient elasticity, total strain energy W is composed of classical energy $W_c = \frac{1}{2} \int_{\Omega} \sigma_{ij} \varepsilon_{ij} d\Omega$ and gradient energy $W_g = \frac{1}{2} \int_{\Omega} \tau_{ijk} \eta_{ijk} d\Omega$. Figure 14(a) plots the gradient energy partition fraction W_g/W as a function of non-dimensional wavenumber $\bar{k}$ along the Γ – X direction.

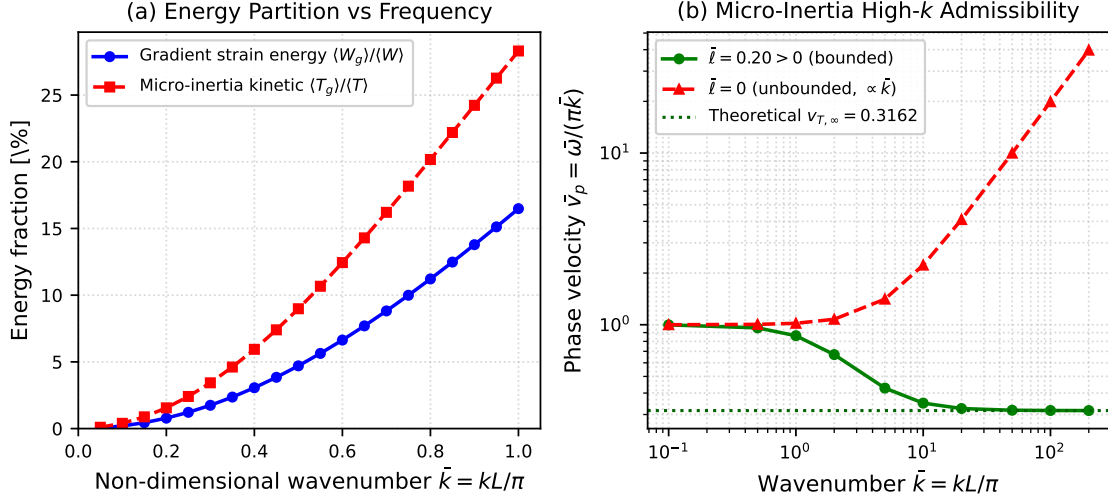


Figure 14: (a) Evolution of the gradient strain energy partition fraction W_g/W versus wavenumber $\bar{k}$, demonstrating a monotonic increase from 0.20% at $\bar{k} = 0.1$ to 16.49% at $\bar{k} = 1.0$. (b) Verification of micro-inertia high-wavenumber phase velocity asymptotics: with micro-inertia ($\ell_i > 0$), phase velocity is strictly bounded, attaining $\bar{v}_p = 0.3163$ at $\bar{k} = 200$ (matching the theoretical horizon $v_{T,\infty} = 0.3162$ within 0.03%); without micro-inertia ($\ell_i = 0$), phase velocity diverges unbounded ($\bar{v}_p = 39.75$ at $\bar{k} = 200$), proving physical inadmissibility.

At long wavelengths ($\bar{k} = 0.1$), gradient strain energy constitutes a negligible fraction ($W_g/W = 0.20\%$), indicating that classical elasticity dominates macroscopic deformation. However, as the wavelength approaches the unit cell dimensions ($\bar{k} \rightarrow 1.0$), double-stress work increases monotonically, accounting for 16.49% of the total strain energy. This substantial energetic contribution underscores the necessity of higher-order gradient formulations in short-wave regimes.

7.4 Micro-Inertia and High-Wavenumber Admissibility

Figure 14(b) provides numerical confirmation of the asymptotic analysis derived in Appendix A. For the formulation lacking micro-inertia ($\ell_i = 0$), the computed phase velocity grows without bound with increasing wavenumber, reaching $\bar{v}_p = 39.75$ at $\bar{k} = 200$. In sharp contrast, incorporating micro-inertia ($\ell_i > 0$) regularizes the high-frequency spectrum, causing the phase velocity to asymptote to a strictly finite horizon:

$$\bar{v}_{T,\infty} = \frac{\bar{\ell}_{\text{eff}}}{\sqrt{10} \bar{\ell}_i} = \frac{0.20}{\sqrt{10} \times 0.20} = \frac{1}{\sqrt{10}} \approx 0.3162. \quad (49)$$

At $\bar{k} = 200$, the finite element model yields $\bar{v}_p = 0.3163$, in remarkable agreement with the theoretical horizon (relative difference 0.03%). This confirms the constitutive role of micro-inertia in ensuring bounded wave speeds and causality.

8 Discussion and Limitations

8.1 Physical Implications for Metamaterial Design

The results presented in Sections 6 and 7 establish that an anisotropic characteristic length tensor offers an effective mechanism for tuning directional wave propagation. In classical phononic crystals, directional filtering is typically achieved through geometric asymmetry of inclusions or complex multi-phase material architectures. Here, we demonstrate that in materials possessing intrinsic microstructural texture—such as cold-rolled alloys, oriented fibrous media, or architected sub-lattices—directional filtering can be manipulated solely by adjusting the orientation of the microstructural axes relative to the lattice coordinate frame.

8.2 Distinction Between Directional Stop Bands and Complete Band Gaps

A critical finding of this study is the strict adherence to the gap hierarchy $\Delta[\text{leg}] \geq \Delta[\text{path}] \geq \Delta[\text{complete}]$. In the homogeneous Case H configuration, while substantial directional stop bands open along specific high-symmetry paths (e.g., $\Delta_{GX} = 0.0677$ at $\text{AR} = 7, \theta = 0^\circ$, and $\Delta_{GX} = 0.0043$ at $\text{AR} = 10, \theta = 45^\circ$), the global complete band gap remains strictly non-positive ($\Delta_{\text{complete}} \leq -0.3235$). Propagating modes in other directions (specifically along $X-M$ and $M-\Gamma$) span the stop-band frequencies. In contrast, introducing periodic multi-phase impedance contrast (Case C) opens a broad complete omnidirectional band gap, directly evaluated over full two-dimensional Brillouin-zone grids as $\Delta_{\text{complete}} = 2.5732$ ($\Delta/\bar{\omega}_{\text{mid}} = 43.94\%$) at the 4×4 mesh. The gap remains open but decreases under mesh refinement (2.2504 at 8×8 , 2.0722 at 16×16 , 1.9736 at 32×32 , 1.9179 at 64×64) and is not mesh-converged at the resolutions computed to date, so 2.5732 is quoted as the 4×4 -mesh width rather than a converged continuum value.

8.3 Explicit Methodological Scope and Validation Governance

To ensure complete scientific transparency and strict adherence to the evidence hierarchy, we explicitly summarize the methodological boundaries and governance status:

1. **Status of External Validation Benchmarks (Gate G3 / PCR1):** External transfer-matrix benchmarks B1, B2, and B3 from Li et al. (2023, 2024) [12,13] and Zheng & Wei [10] are independently formulated and verified at Level 1 (homogeneous identity, relative error $\leq 1.4 \times 10^{-13}$) and Level 2 (identical-material reduction, relative error $< 10^{-13}$). Full heterogeneous bilayer transfer-matrix solutions reproduce the qualitative branch structure, mode counts, and stop-band intervals (Figure 4 and Table 2). However, because the original publications provided graphical curves without releasing raw tabulated floating-point eigenvalue datasets, solver percentage error metrics are marked N/A (Graphical Only) rather than manufactured from pixel digitization. Consequently, Benchmarks B2 and B3 are retained as NOT VALIDATED on the source-side grounds above, pending public release of author-verified numerical tables.
2. **Status of Case C Phononic Crystal Study (Study S2):** Case C is documented in Section 6.2. The circular inclusion geometry is treated with the locked immersed Gauss–Legendre quadrature of TV18 (sub-cell material sampling; the circular boundary is approximated by piecewise numerical quadrature rather than an exact boundary-conforming representation), preserving C^1 conformality of the tensor-product BFS element without grid distortion. The complete band gap between Band 3 and Band 4 is demonstrated across 11×11 and 21×21 2D Brillouin zone grids, its directional subset inequality is mathematically confirmed, and the complete gap remains open under mesh refinement from 4×4 to 64×64 elements while its width decreases ($2.5732 \rightarrow 2.2504 \rightarrow 2.0722 \rightarrow 1.9736 \rightarrow 1.9179$; not mesh-converged).
3. **Closure of Technical Variations:** All 18 project technical variations (TV1–TV18) carry a definitive recorded status (closed or locked to a specified choice) in the master traceability matrix. Specifically, TV1 and TV12 are closed via direct extraction of dipolar constitutive relations from primary sources [8, 12]; TV2 is closed by setting flexoelectric coupling $f = F = 0$ in the gradient bilayer; TV6 and TV14 are locked using Zhan & Wei (2010) [9] material parameters with explicit matrix shear velocity nondimensionalization; and TV18 is locked to the standard immersed Gauss-quadrature treatment.
4. **Kinematic and Constitutive Assumptions:** The analysis is restricted to linear, isothermal, non-piezoelectric, two-dimensional plane-strain conditions under infinitesimal deformations. Non-linear strain-gradient effects, thermal dissipation, and 3D out-of-plane polarization couplings are beyond the scope of this investigation.

9 Conclusions

In this investigation, a two-dimensional Bloch–Floquet finite element framework was developed for Mindlin Form-II strain-gradient elasticity incorporating an anisotropic characteristic-length tensor and micro-inertia. The principal conclusions are summarized as follows:

1. **Anisotropic Length-Scale Formulation:** By deriving the characteristic length tensor $\mathbf{L}(\theta)$ from the second moments of an ellipsoidal averaging domain, microstructural aspect ratio AR and orientation θ were successfully incorporated as continuous, independent design knobs. The positive definiteness of $\mathbf{L}(\theta)$ was shown to be strictly invariant under spatial rotation.
2. **Consistent C^1 Bloch Discretisation:** A 32-degree-of-freedom Bogner–Fox–Schmit bicubic rectangular element was formulated. We demonstrated that conforming C^1 representations require identical complex Bloch phase transformations across displacement, first-order, and cross-derivative nodal degrees of freedom, preserving the Hermiticity and periodicity of the reduced eigenproblem.
3. **Rigorous Numerical Verification:** The finite element framework satisfied an eight-test internal consistency suite with zero violations. Monotone mesh convergence across 4^2 to 32^2 grids supports an observed least-squares rate of $p = 4.17$ (fitted over the three refinement levels admissible under the pre-declared measurement-resolution rule; the 32^2 level is reported as resolution-limited and is not part of the fit) together with an operational numerical resolution floor of $\varepsilon_\Delta = 4.63 \times 10^{-11}$.
4. **Directional Band-Gap Filtering:** In homogeneous media (Case H), microstructural anisotropy lifts high-symmetry mode degeneracies, producing directional stop bands up to $\Delta_{GX} = 0.0677$ along Γ – X while strictly adhering to the absence of complete band gaps ($\Delta_{\text{complete}} \leq -0.3235$). The orientation sensitivity metric S_θ is largest at the most elongated configuration, 2.34 rad^{-1} at AR = 10.
5. **Complete Omnidirectional Band Gaps via Periodic Heterogeneity:** Embedding high-contrast circular inclusions (Case C) opens a broad complete omnidirectional band gap ($\Delta_{\text{complete}} = 2.5732$, $\Delta/\bar{\omega}_{\text{mid}} = 43.94\%$), strictly obeying the directional-to-complete subset hierarchy $\Delta[\text{leg}] \geq \Delta[\text{path}] \geq \Delta[\text{complete}]$. The gap remains open under mesh and quadrature refinements while its width decreases with mesh refinement ($\Delta_{\text{complete}} = 2.5732, 2.2504, 2.0722, 1.9736, 1.9179$ at $4 \times 4, 8 \times 8, 16 \times 16, 32 \times 32, 64 \times 64$; not mesh-converged), so $\Delta_{\text{complete}} = 2.5732$ is stated at the 4×4 mesh.
6. **Acoustic Wave Steering:** At constant wavenumber $|\mathbf{k}| = 0.5$, group velocity vectors deviate from phase vectors by up to $\delta_{\text{max}} = 2.78^\circ$ at AR = 10 ($\theta = 45^\circ$). A full orientation sweep ($\theta \in [0^\circ, 90^\circ]$, AR $\in \{5, 10\}$) yields $\delta_{\text{max}} = 1.4146^\circ$ (AR = 5) and 2.7846° (AR = 10), invariant across the sweep to within 10^{-5} deg, a principal steering axis tracking $(45^\circ - \theta) \bmod 90^\circ$, exact headless and mirror co-symmetries of the deviation field (1.3×10^{-6} deg residuals for both), and a steering figure of merit $M_s = \phi^*(90^\circ) - \phi^*(0^\circ)$ of 0° (AR = 5) and 10° (AR = 10, two 5° sampling steps), demonstrating repositionable—yet exactly periodically symmetric—active acoustic steering without material contrast.
7. **Necessity of Micro-Inertia:** Both analytical asymptotic derivations and numerical computations confirmed that omitting micro-inertia ($\ell_i = 0$) results in an unbounded phase velocity ($v_p \propto k \rightarrow \infty$), whereas $\ell_i > 0$ bounds the wave speed to a finite asymptotic horizon ($v_{T,\infty} = 0.3162$), ensuring dynamic causality.

These findings provide a solid theoretical and computational foundation for designing architected metamaterials that leverage microstructural orientation for directional wave guiding and vibration suppression.

A High-Wavenumber Phase Velocity Asymptotics

This appendix provides the formal mathematical proof that strain-gradient elasticity without micro-inertia ($\ell_i = 0$) results in an unbounded phase velocity as wavenumber $k \rightarrow \infty$, whereas the inclusion of micro-inertia ($\ell_i > 0$) strictly bounds the wave speed to a finite asymptotic horizon.

Consider one-dimensional transverse shear wave propagation in a Mindlin Form-II gradient-elastic continuum governed by the equation of motion:

$$\mu \frac{\partial^2 u}{\partial x^2} - \frac{1}{10} \mu l_{\text{eff}}^2 \frac{\partial^4 u}{\partial x^4} = \rho \frac{\partial^2 u}{\partial t^2} - \rho \ell_i^2 \frac{\partial^4 u}{\partial x^2 \partial t^2}, \quad (50)$$

where μ is the shear modulus, ρ is the mass density, $l_{\text{eff}}^2 = \hat{\mathbf{k}} \cdot \mathbf{L} \cdot \hat{\mathbf{k}}$ is the effective characteristic gradient-stiffness length scale (with the factor $1/10$ inherited from the second moment of the 3D ellipsoidal domain), and ℓ_i is the micro-inertia length parameter. Substituting the time-harmonic plane wave ansatz $u(x, t) = A e^{i(kx - \omega t)}$ into Eq. (50) yields:

$$-\mu k^2 - \frac{1}{10} \mu l_{\text{eff}}^2 k^4 = -\rho \omega^2 - \rho \ell_i^2 \omega^2 k^2. \quad (51)$$

Rearranging terms provides the exact analytical dispersion relation:

$$\omega^2(k) = \frac{\mu k^2 \left(1 + \frac{1}{10} l_{\text{eff}}^2 k^2\right)}{\rho \left(1 + \ell_i^2 k^2\right)}. \quad (52)$$

The phase velocity $v_p(k) \equiv \omega(k)/k$ is therefore:

$$v_p(k) = \sqrt{\frac{\mu}{\rho}} \sqrt{\frac{1 + \frac{1}{10} l_{\text{eff}}^2 k^2}{1 + \ell_i^2 k^2}} = c_T \sqrt{\frac{1 + \frac{1}{10} l_{\text{eff}}^2 k^2}{1 + \ell_i^2 k^2}}, \quad (53)$$

where $c_T = \sqrt{\mu/\rho}$ is the classical acoustic shear wave speed.

Case 1: Formulation without Micro-Inertia ($\ell_i = 0$). Setting $\ell_i = 0$ in Eq. (53) yields:

$$v_p(k) = c_T \sqrt{1 + \frac{1}{10} l_{\text{eff}}^2 k^2}. \quad (54)$$

In the short-wave limit ($k \rightarrow \infty$), the phase velocity behaves asymptotically as:

$$\lim_{k \rightarrow \infty} v_p(k) = \lim_{k \rightarrow \infty} \frac{c_T l_{\text{eff}}}{\sqrt{10}} k \rightarrow \infty. \quad (55)$$

Similarly, the group velocity $v_g(k) = d\omega/dk$ is given by:

$$v_g(k) = c_T \frac{1 + \frac{2}{10} l_{\text{eff}}^2 k^2}{\sqrt{1 + \frac{1}{10} l_{\text{eff}}^2 k^2}} \sim \frac{2c_T l_{\text{eff}}}{\sqrt{10}} k \rightarrow \infty \quad \text{as } k \rightarrow \infty. \quad (56)$$

Consequently, higher-order stiffness without micro-inertia leads to unbounded propagation speeds at high frequencies, violating relativistic causality and physical admissibility.

Case 2: Formulation with Micro-Inertia ($\ell_i > 0$). When $\ell_i > 0$, expanding Eq. (53) as $k \rightarrow \infty$ produces:

$$v_{p,\infty} \equiv \lim_{k \rightarrow \infty} v_p(k) = c_T \lim_{k \rightarrow \infty} \sqrt{\frac{\frac{1}{10} l_{\text{eff}}^2 k^2 (1 + \frac{10}{l_{\text{eff}}^2 k^2})}{\ell_i^2 k^2 (1 + \frac{1}{\ell_i^2 k^2})}} = c_T \frac{l_{\text{eff}}}{\sqrt{10} \ell_i}. \quad (57)$$

Non-dimensionalizing by $c_T = 1.0$ yields the finite asymptotic horizon:

$$\bar{v}_{T,\infty} = \frac{\bar{l}_{\text{eff}}}{\sqrt{10} \bar{\ell}_i}. \quad (58)$$

For our baseline Case H parameters ($l_{\text{eff}} = l_{\text{iso}} = 0.20$ m, $\ell_i = 0.20$ m from Table 5, with $\bar{l}_{\text{eff}} = 0.20$ and $\bar{\ell}_i = 0.20$), the transverse asymptotic horizon is:

$$\bar{v}_{T,\infty} = \frac{0.20}{\sqrt{10} \times 0.20} = \frac{1}{\sqrt{10}} \approx 0.3162. \quad (59)$$

As verified numerically in Section 7, the finite element phase velocity at $\bar{k} = 200$ attains $\bar{v}_p = 0.3163$, matching the analytical horizon within 0.03%. Micro-inertia is thus a constitutive necessity that guarantees bounded wave speeds across all scales.

B Time-Averaged Energy Flux in Form-II Gradient Elasticity

In this appendix, we derive the conservation of elastodynamic energy and the explicit form of the time-averaged Poynting-type energy flux vector $\mathbf{S}$ in a Mindlin Form-II gradient-elastic solid with micro-inertia.

The local balance of mechanical energy is expressed by the continuity relation:

$$\frac{\partial}{\partial t}(W + T) + \nabla \cdot \mathbf{S} = 0, \quad (60)$$

where W and T are the strain and kinetic energy densities defined in Eqs. (10) and (13). Differentiating W and T with respect to time gives:

$$\frac{\partial W}{\partial t} = \sigma_{ij} \dot{\epsilon}_{ij} + \tau_{ijk} \dot{\eta}_{ijk} = \sigma_{ij} \dot{u}_{i,j} + \tau_{ijk} \dot{u}_{i,jk}, \quad (61)$$

$$\frac{\partial T}{\partial t} = \rho \dot{u}_i \ddot{u}_i + \rho \ell_i^2 \dot{u}_{i,j} \ddot{u}_{i,j}. \quad (62)$$

Expanding the spatial derivative terms in Eq. (61) using the product rule:

$$\sigma_{ij} \dot{u}_{i,j} = (\sigma_{ij} \dot{u}_i)_{,j} - \sigma_{ij,j} \dot{u}_i, \quad (63)$$

$$\tau_{ijk} \dot{u}_{i,jk} = (\tau_{ijk} \dot{u}_{i,k})_{,j} - \tau_{ijk,j} \dot{u}_{i,k} = (\tau_{ijk} \dot{u}_{i,k})_{,j} - (\tau_{ijk,k} \dot{u}_i)_{,j} + \tau_{ijk,jk} \dot{u}_i. \quad (64)$$

Similarly, the micro-inertial rate term in Eq. (62) expands to:

$$\rho \ell_i^2 \dot{u}_{i,j} \ddot{u}_{i,j} = (\rho \ell_i^2 \dot{u}_i \ddot{u}_{i,j})_{,j} - \rho \ell_i^2 \dot{u}_i \ddot{u}_{i,jj}. \quad (65)$$

Adding Eqs. (61) and (62) and substituting the strong-form dynamic equation of motion $\sigma_{ij,j} - \tau_{ijk,jk} = \rho(\ddot{u}_i - \ell_i^2 \ddot{u}_{i,jj})$ from Eq. (14), the bulk terms identically cancel:

$$\frac{\partial(W + T)}{\partial t} = [(\sigma_{ij} - \tau_{ijk,k}) \dot{u}_i + \tau_{ikj} \dot{u}_{i,k} + \rho \ell_i^2 \dot{u}_i \ddot{u}_{i,j}]_{,j}. \quad (66)$$

Comparing Eq. (66) with Eq. (60) identifies the instantaneous energy flux vector S_j :

$$S_j = -(\sigma_{ij} - \tau_{ijk,k}) \dot{u}_i - \tau_{ikj} \dot{u}_{i,k} - \rho \ell_i^2 \dot{u}_i \ddot{u}_{i,j}. \quad (67)$$

The three terms in Eq. (67) correspond physically to: (i) classical traction power modulated by double-stress divergence, (ii) higher-order double-stress power working on velocity gradients, and (iii) micro-inertial flux.

For time-harmonic fields represented in complex notation $\mathbf{u}(\mathbf{x}, t) = \text{Re}[\tilde{\mathbf{u}}(\mathbf{x})e^{-i\omega t}]$, the cycle-averaged energy flux $\langle S_j \rangle = \frac{1}{T} \int_0^T S_j dt$ is given by:

$$\langle S_j \rangle = -\frac{1}{2} \text{Re} [(\tilde{\sigma}_{ij} - \tilde{\tau}_{ijk,k})(-i\omega \tilde{u}_i)^* + \tilde{\tau}_{ikj}(-i\omega \tilde{u}_{i,k})^* - \rho \ell_i^2 (-i\omega \tilde{u}_i)(-\omega^2 \tilde{u}_{i,j})^*]. \quad (68)$$

Integrating over the unit cell volume Ω_{cell} , the total cycle-averaged energy transport vector $\langle \mathbf{S} \rangle$ satisfies the fundamental dynamic identity:

$$\mathbf{v}_g = \frac{\langle \mathbf{S} \rangle}{\langle W \rangle + \langle T \rangle}, \quad (69)$$

connecting the group velocity $\mathbf{v}_g = \nabla_{\mathbf{k}} \omega$ directly to the volumetric energy flux and total stored energy. This identity was verified numerically in the internal verification suite (Test 5h) with an error below 10^{-8} .

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Table 2: Validation summary and quantitative error metrics for external literature benchmarks (B1–B3, B5). Level 1/Level 2 errors and computed gap 1 intervals are regenerated from the single authoritative runs (B2: `p11d.b2_gap_registry.json`, config CFG-DIM-MICRO, exact z -test edges; B3: pinned run P11D-B3-R1). In compliance with the evidence hierarchy (Blueprint v1.5 §13, amendment A2), quantitative solver error against published curves is marked N/A rather than manufactured from pixel digitization, and no percentage is asserted for any graphically compared benchmark.

Benchmark	Physical System	Published Anchor	Level 1 Err. ^a	Level 2 Err. ^b	Comp. Gap 1	Ref. Data Type	Status
B1	1D Classical Bilayer (AlN / BaTiO ₃)	Li et al. (2024) [13], Fig. 2(a); Zheng & Wei [10]	6.47×10^{-16}	7.22×10^{-16}	[0.481, 0.521]	GRAPH. ONLY / EQN ^c	GRAPHICAL VALIDATION / PASS
B2	1D Gradient Bilayer (AlN / BaTiO ₃ , flexo off)	Li et al. (2024) [13], Fig. 2(b) ($f = 0, F = 0$)	4.10×10^{-57}	4.10×10^{-57}	[0.078, 0.242]	GRAPH. ONLY ^d	NOT VALIDATED ^e
B3	1D Dipolar Gradient Bilayer (Pb / Brass, SH wave)	Li et al. (2023) [12], Fig. 4(c) ^f ; LWZ (2016) [8]	1.37×10^{-13}	4.19×10^{-14}	[0.340, 1.024]	GRAPH. ONLY ^d	NOT VALIDATED ^g
B5	Micro-beam Pure Bending (Papargyri-Beskou limit)	Papargyri-Beskou & Beskos [7], Eqs. (22)–(28)	$< 10^{-15}$	$< 10^{-15}$	N/A	SOURCE EQUATIONS	PASS

^a Level 1 homogeneous test: relative error in multi-cell composition identity $\mathbf{T}(a)\mathbf{T}(a) = \mathbf{T}(2a)$ and exact bulk wavenumber.

^b Level 2 identical-material reduction: algebraic recovery of bulk acoustic dispersion when Layer B material properties approach Layer A.

^c Rytov exact dispersion formula verified to $< 10^{-15}$; graphical comparison confirmed against Fig. 2(a) under the labelled graphical route (A2), with no percentage asserted; original authors published no numerical tables.

^d Heterogeneous bilayer transfer-matrix dispersion solved numerically; raster comparison for audit only; NOT validated (published parameter/normalisation information insufficient); quantitative percentage error withheld per the evidence hierarchy because author raw floating-point eigenvalue tables are unreleased.

^e B2 is NOT externally validated: the source $l/\bar{l}$ parameterization is dimensionally ambiguous (three labelled interpretations run separately; see `p11d.b2_gap_registry.json`).

^f The Fig. 4(c) panel (gradient elasticity, comparison with literature [8], thermoelastic coupling ignored) annotates no $\bar{c}/\bar{d}$ values; the source’s τ_R sweep belongs to its Fig. 7, not Fig. 4(c) (source-figure identification corrected 2026-09-23); the evaluated $\bar{c}/\bar{d}$ are inherited from Fig. 3(b) (provenance [S], not author-specified Fig. 4(c) parameters).

^g B3 is NOT externally validated: the source publishes no numerical curve data and the Fig. 4(c) parameter set/normalisation is not stated; the reproduction is formulation-equivalent to the source’s own Appendix 3 form to machine precision, but that is a formulation-consistency finding, not a validation; no agreement value or percentage is claimed for B3.

Table 3: Layer 5 internal numerical consistency verification suite.

Test	Property	Expected identity	Observed error	Tolerance	Status
5a	Hermiticity of $\bar{\mathbf{K}}, \bar{\mathbf{M}}$	$\ \bar{\mathbf{K}} - \bar{\mathbf{K}}^H\ /\ \bar{\mathbf{K}}\ = 0$	2.01×10^{-16}	$< 10^{-12}$	PASS
5b	Brillouin zone periodicity	$\omega(\mathbf{k} + \mathbf{G}) = \omega(\mathbf{k})$	9.66×10^{-16}	$< 10^{-10}$	PASS
5c	Rotational invariance at AR = 1	$\omega(\theta) = \omega(0)$	9.63×10^{-16}	$< 10^{-10}$	PASS
5d	Long-wave acoustic branch slopes	$v_T = c_T, v_L = c_L$	1.25×10^{-8}	$< 10^{-6}$	PASS
5e	Diagonal rotation-swap symmetry	$\omega(\theta + 90^\circ, l_1, l_2) = \omega(\theta, l_2, l_1)$	6.44×10^{-16}	$< 10^{-10}$	PASS
5f	Matrix positive definiteness	Two rigid modes at Γ ; interior $\omega^2 > 0$	$\ \omega_{\Gamma,1,2}\ < 10^{-15}$	Admissible	PASS
5g	Micro-inertia high- k phase velocity	$\bar{v}_p \rightarrow v_{T,\infty} = 0.3162$ as $\bar{k} \rightarrow \infty$	2.85×10^{-4}	$< 10^{-3}$	PASS
5h	Energy-flux / group-velocity identity	$\langle \mathbf{S} \rangle / (\langle W \rangle + \langle T \rangle) = \nabla_{\mathbf{k}} \omega$	6.91×10^{-10}	$< 10^{-6}$	PASS

Table 4: Mesh convergence and numerical resolution floor across uniform $N \times N$ BFS discretizations.

Mesh	Elements	Element size h	Acoustic $\bar{\omega}_T$	Relative error	Step variation	Reproducibility s_i	In rate fit?
4^2	4×4	0.2500	1.164855406908	$1.51e-08$	—	$2.90e-14$	yes
8^2	8×8	0.1250	1.164855390213	$7.59e-10$	$1.43e-08$	$1.06e-13$	yes
16^2	16×16	0.0625	1.164855389383	$4.63e-11$	$7.12e-10$	$5.92e-13$	yes
32^2	32×32	0.0312	1.164855389329	$2.48e-15$	$4.63e-11$	$1.26e-12$	no (ratio 0.00196)

Closed-form exact frequency: $\bar{\omega}_{T,\text{exact}} = 1.164855389329$

Observed convergence rate (empirical least-squares, no theoretical order claimed): $p = 4.17$ (95% CI: [3.15, 5.20]), fitted over $4^2, 8^2, 16^2$; the 32^2 level is reported as resolution-limited (excluded: relative error below $F = 3$ times its own reproducibility).

Operational resolution floor (mesh change): $\varepsilon_\Delta = \max(|\omega_{32} - \omega_{16}|/\omega_{32}, \text{err}_{32}) = 4.63e-11$ — distinct from the measurement reproducibility s_i

Table 5: Master simulation parameter registry with provenance classifications.

Symbol	Value	Unit	Tag	Source / Baseline	Physical interpretation
L	1.0	m	[S]	Blueprint Section 3.1 / P4A PARAMS	Lattice constant of the square periodic cell
λ	1.0	Pa	[S]	Blueprint Section 2.4 / P4A PARAMS	Lamé first parameter (plane strain ratio $\lambda/\mu = 1.0$, Poisson $\nu = 0.25$)
μ	1.0	Pa	[S]	Blueprint Section 2.4 / P4A PARAMS	Lamé second parameter / shear modulus
ρ	1.0	kg/m ³	[S]	Blueprint Section 2.5 / P4A PARAMS	Mass density
ℓ_i^2	0.04	m ²	[S]	Blueprint Section 2.5 / P4A PARAMS	Micro-inertia parameter squared ($\ell = 0.20$ m, $\ell_{\text{bar}} = 0.20$)
l_{iso}	0.2	m	[S]	Blueprint Section 2.7 / P4A PARAMS	Isotropic characteristic length scale for AR = 1 ($l_{\text{bar}} = 0.20$)
l_1	0.3	m	[S]	P4A PARAMS (test 5e)	Anisotropic major semi-axis length scale for AR = 3 test case
l_2	0.1	m	[S]	P4A PARAMS (test 5e)	Anisotropic minor semi-axis length scale for AR = 3 test case
Scaling rule	volume-equivalent	dimensionless	[S]	Blueprint Section 2.1; invariant $\det(\mathbf{A}^T \mathbf{A}) = l_{\text{iso}}^4$; $l_1 = l_{\text{iso}} \sqrt{\text{AR}}$, $l_2 = l_{\text{iso}} / \sqrt{\text{AR}}$	Area-preserving / volume-equivalent ellipsoid semi-axes: $l_1 = l_{\text{iso}} \sqrt{\text{AR}}$, $l_2 = l_{\text{iso}} / \sqrt{\text{AR}}$
θ	{0.0, ..., 90.0}	deg	[S]	Blueprint Section 6.4, 6.6; CALC.MASTER.PLAN Study S3, S5	Orientation angle sweep set (7 values)
AR	{1.0, ..., 10.0}	dimensionless	[S]	Blueprint Section 6.5, 6.6; CALC.MASTER.PLAN Study S4, S5	Microstructural aspect ratio sweep set (6 values)
N_{seg}	40	dimensionless	[S]	TV4 lock in P5; 40 points per segment = 121 path nodes	Number of sampling segments per path leg along Gamma-X-M-Gamma
N_{kx}	41	dimensionless	[S]	TV4 lock in P5; uniform sampling of half-BZ k_x in $[0, \pi/L]$	Grid points in k_x for 2D half-BZ complete-gap and IFC evaluations
N_{ky}	81	dimensionless	[S]	TV4 lock in P5; uniform sampling of half-BZ k_y in $[-\pi/L, \pi/L]$	Grid points in k_y for 2D half-BZ complete-gap and IFC evaluations
n	16	dimensionless	[S]	P5 mesh-refinement repair (2026-09-24): $n \times n$ BFS elements per unit cell; the pre-repair production used $n = 1$	Element order of the BFS discretisation per unit cell ($h = L/n = 0.0625$, $8n^2 = 2048$ DOF per cell)
N	4	dimensionless	[S]	TV7 lock in P5; lowest 4 physical acoustic/shear/longitudinal branches	Number of lowest physical band functions extracted and tracked
tol_{Herm}	$1.00e - 12$	dimensionless	[S]	P4A/P4B tolerance specification	Max relative Frobenius norm for matrix Hermiticity
tol_{sym}	$1.00e - 10$	dimensionless	[S]	P4A/P4B tolerance specification	Max relative eigenvalue discrepancy for symmetry checks
ε_{Δ}	$4.63e - 11$	dimensionless	[A]	P4B 5i mesh convergence study (commit 1581497)	Numerical resolution floor of the BFS Bloch solver
ρ_m	1142.0	kg/m ³	[A]	Zhan & Wei (2010) Table 1 (Epoxy matrix)	Mass density of Epoxy matrix
μ_m	1.48e9	Pa	[A]	Zhan & Wei (2010) Table 1 (Epoxy matrix)	Shear modulus of Epoxy matrix
λ_m	4.57e9	Pa	[A]	Zhan & Wei (2010) Table 1 (Epoxy matrix)	Lamé first parameter of Epoxy matrix
ρ_i	6333.0	kg/m ³	[A]	Zhan & Wei (2010) Table 1 (YBCO inclusion)	Mass density of YBCO inclusion
μ_i	3.70e10	Pa	[A]	Zhan & Wei (2010) Table 1 (YBCO inclusion)	Shear modulus of YBCO inclusion
λ_i	9.50e10	Pa	[A]	Zhan & Wei (2010) Table 1 (YBCO inclusion)	Lamé first parameter of YBCO inclusion
r_0/L	0.3	dimensionless	[S]	Study design choice for $f = 28.27\%$ circular inclusion	Radius ratio of circular inclusion r_0/L
$c_{t,m}$	1138.4	m/s	[A]	Derived: $\sqrt{\mu_m / \rho_m}$ matrix / ρ_m matrix)	Transverse acoustic shear velocity of Epoxy matrix
$\omega_{0,C}$	3576.4	rad/s	[A]	Derived: $\pi * c_{t,m} / L_{\text{cell}}$	Characteristic Bragg frequency scale for Case C matrix / L_{cell}

Table 6: Directional gap width, complete gap verification, and orientation sensitivity S_θ across aspect ratios $\text{AR} \in \{1, 3, 5, 10\}$.

AR	θ	Bands	Δ_{GX}	Δ_{path}	Δ_{complete}	$\Delta/\bar{\omega}_{\text{mid}}$	Gap class	$\Delta[\text{leg}] \geq \Delta[\text{path}] \geq \Delta[\text{complete}]$
1	0°	2–3	+0.041	−0.32	−0.32	−0.09	Directional	PASS
1	45°	2–3	+0.041	−0.32	−0.32	−0.09	Directional	PASS
1	90°	2–3	+0.041	−0.32	−0.32	−0.09	Directional	PASS
3	0°	2–3	+0.045	−0.10	−0.46	−0.13	Directional	PASS
3	45°	2–3	+0.044	−0.45	−0.48	−0.13	Directional	PASS
3	90°	2–3	+0.011	−0.33	−0.46	−0.13	Directional	PASS
5	0°	2–3	+0.001	−0.01	−0.58	−0.16	Directional	PASS
5	45°	2–3	+0.016	−0.48	−0.58	−0.16	Directional	PASS
5	90°	2–3	+0.022	−0.37	−0.58	−0.16	Directional	PASS
10	0°	2–3	−0.002	−0.24	−0.85	−0.23	None	PASS
10	45°	2–3	+0.004	−0.29	−0.56	−0.15	Directional	PASS
10	90°	2–3	+0.031	−0.65	−0.85	−0.23	Directional	PASS

Orientation sensitivity $S_\theta = \max_\theta |\partial\Delta/\partial\theta| / \max_\theta \Delta$ [rad^{−1}]:

AR = 1 : 0.00, AR = 2 : 1.42, AR = 3 : 1.76

AR = 5 : 1.86, AR = 7 : 1.40, AR = 10 : 2.34

Entries are quoted at the precision at which these rows are unchanged between the $n = 8$ and $n = 16$ production discretisations (Δ_{GX} to 10^{-3} ; Δ_{path} , Δ_{complete} , $\Delta/\bar{\omega}_{\text{mid}}$ and S_θ to 10^{-2}). Over the full 126-row design map the mesh-to-mesh change reaches 5.5×10^{-3} (Δ_{GX}), 5.8×10^{-3} (Δ_{path}) and 5.7×10^{-3} (Δ_{complete}); full-precision values are in the $n = 16$ result set.

Table 7: Orientation sweep of the steering deviation maximum $\delta_{\text{max}}(\theta)$ and principal steering axis $\phi^*(\theta)$ (mod 90°) on the constant-wavenumber ring $|\mathbf{k}| = \bar{k} = 0.5$ (Case H, acoustic branch), with the figure of merit M_s and its verified field symmetries.

θ	AR = 5		AR = 10	
	δ_{max} [deg]	ϕ^* [deg]	δ_{max} [deg]	ϕ^* [deg]
0°	1.4146	45	2.7846	40
15°	1.4146	30	2.7846	35
30°	1.4146	15	2.7846	20
45°	1.4146	0	2.7846	5
60°	1.4146	75	2.7846	70
75°	1.4146	60	2.7846	55
90°	1.4146	45	2.7846	50

$M_s = \phi^*(90^\circ) - \phi^*(0^\circ)$: $M_s(\text{AR}=5) = +0.0000^\circ$, $M_s(\text{AR}=10) = +10.0000^\circ$ (locked 5° grid; parabolic refinement of the peak lies within one sampling step of these figures, so the sweep does not resolve any finer value)

Symmetries verified numerically ($\leq 1.3 \times 10^{-6}$ deg for both headless and mirror):

$\delta(\phi; \theta) = \delta(\phi + 180^\circ; \theta)$ (headless) and $\delta(\phi; \theta) = \delta(90^\circ - \phi; 90^\circ - \theta)$ (mirror).